SAMCE-CR: SAR-Guided Alignment and Multifrequency Collaborative Enhancement for Cloud Removal in Remote Sensing Images Fusion

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***Abstract*—In global Earth observation, multispectral imaging is frequently hindered by extensive cloud cover, leading to observation gaps. Introducing synthetic aperture radar (SAR) data with all-weather penetration capability for multimodal cloud removal has become a key approach to achieving global spatiotemporal seamless remote sensing monitoring. Existing SAR-optical fusion methods often lack effective decoupling and differentiated processing of heterogeneous data. This leads to texture distortion in reconstructed images, limiting the accuracy of downstream interpretation and target recognition. Therefore, a cloud removal method based on SAR-guided alignment and multifrequency collaborative enhancement is proposed in this article. First, the spatial geometric features of SAR images are mined by multireceptive field gating mechanism, and robust structural priors are extracted for under-cloud ground object reconstruction while effectively suppressing coherent speckle noise; Second, with the help of the deformable alignment module, the geometric alignment of SAR and optical images is realized with reference to optical images, alleviating the problem of scale and spatial misalignment; Finally, through the multifrequency collaborative enhancement module, the high and low frequency information are adaptively separated, and an improved attention mechanism is adopted to enhance the high and low frequency information, respectively, which effectively suppresses cloud interference and maintains surface details. Results on the M3R-CR and LuojiaSET-OSFCR datasets show that the proposed method comprehensively outperforms the other eight compared methods. Compared with the suboptimal method, the peak signal-to-noise ratios (PSNRs) achieved by the proposed method on the two datasets are improved by 0.5643 and 0.1964 dB, respectively. The source code of SAMCE-CR is shared at https://github.com/RSIDEA-ECUT/SAMCE-CR**

***Index Terms*—Cloud removal, fusion, multifrequency collaborative enhancement, optical remote sensing image.**

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I. Introduction

With its abundant surface spectral information, optical remote sensing images play an important role in the core tasks of global Earth observation, resource dynamic assessment, and disaster emergency response [1], [2]. However, due to complex and changeable atmospheric conditions, widespread cloud cover around the world hinders effective observation by optical sensors all year round. This inevitable cloud occlusion not only causes a lack of information about large-scale ground objects in the spatial dimension but also leads to the discontinuity of Earth observation data in the spatial and temporal dimensions [3], [4]. Insufficient data directly undermines the effective utilization of optical remote sensing imagery and fundamentally compromises the reliability of downstream applications, including high-precision ground object interpretation and target recognition [2]. Therefore, how to efficiently restore the real features of cloud-covered areas and reconstruct high-quality cloud-free images to ensure continuous and reliable Earth observation has become an important research problem in the field of remote sensing image processing [5], [6].

To solve this problem, researchers have proposed a variety of solutions, primarily encompassing spatial inference and temporal reference-based methods [7], [8]. Spatial inference-based methods attempt to infer cloud-covered content from uncovered regions within a single image; however, this approach often yields blurring and artifacts when confronted with extensive thick cloud cover or complex ground object textures [9]. Temporal reference-based methods use historical cloud-free images of the same location as a reference. Although performance is improved, it is highly dependent on

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the availability of cloud-free data, resulting in limited applicability in scenarios with long-term cloudiness or drastic surface changes. In general, the aforementioned schemes are constrained by the physical limitations inherent to a single optical modality, while depending excessively upon either local spatial information contained within the image or historical auxiliary information across the temporal dimension. To transcend the limitations of optics, the integration of cross-modal information has emerged as a promising new paradigm [10], [11]. With its unique microwave imaging mechanism, synthetic aperture radar (SAR) has all-weather imaging capability that penetrates clouds and is not affected by meteorological conditions [12], [13]. Attributable to this characteristic, the backscattering information acquired by SAR can provide the essential physical basis for reconstructing the surface structure obscured by cloud cover [14]. However, as SAR and optical images constitute typical heterogeneous data modalities, the direct utilization of SAR for cloud removal assistance encounters substantial cross-modal discrepancy challenges, primarily attributable to the absence of spectral information in SAR data, coupled with inevitable speckle noise contamination and geometric distortion artifacts. Although existing fusion methods have achieved significant progress in cross-modal feature extraction, deficiencies persist in spatially accurate alignment and deep frequency-domain complementarity within complex scenes, resulting in local structural blurring or spectral distortion in reconstructed images.

To address the above challenges, this article proposes an SAR-guided alignment and multifrequency collaborative enhancement method for cloud removal, designated as SAMCE-CR. To verify the effectiveness of the proposed method, comparative experiments were conducted against eight state-of-the-art methods on the M3R-CR and LuojiaSET-OSFCR datasets, with performance assessed using six evaluation metrics. The main contributions of this study are as follows.

1) *Multireceptive Structural Prior Module:* To address the challenges of coherent speckle noise and scale variations in SAR imagery, this module constructs parallel multireceptive-field feature streams to adaptively capture multilevel spatial geometric information. Simultaneously, a content-aware gating mechanism is employed to effectively distinguish object edges from high-frequency noise, thereby enabling adaptive feature extraction. Coupled with a residual up-sampling architecture, this module provides robust structural prior guidance for cross-modal fusion while concurrently aligning spatial resolution, thereby attenuating the uncertainty inherent in under-cloud ground object reconstruction.

2) *Deformable Cross-Modal Alignment Module:* Facing the scale and geometric discrepancies between SAR and optical imagery, this module establishes a deformable sampling architecture steered by optical features, which adaptively predicts spatial offsets and reconfigures the sampling lattice points within local neighborhoods. Concurrently, a spatial and channel attention mechanism is employed to effectively filter redundant information during the alignment process, thereby enabling fine-grained geometric interaction between cross-modal features. Combined with a dynamic offset matching design, this structure not only mitigates artifact propagation and texture misalignment induced by spatial discrepancies but also furnishes high-precision, pixel-level correspondence for subsequent under-cloud ground object reconstruction.

3) *Multifrequency Domain Synergistic Gain Module:* To solve the frequency-domain aliasing degradation induced by cloud layers, this module constructs an adaptive dual-frequency feature stream based on the fast Fourier transform (FFT), enabling precise decoupling of high- and low-frequency sub-band information. Concurrently, a channel-space-cross-attention mechanism is employed to organize cross-frequency domain features, which effectively suppresses low-frequency cloud layer interference while accentuating high-frequency details of authentic ground objects, thereby achieving differentiated frequency feature extraction. Combined with inverse FFT (iFFT) reconstruction and a spatially adaptive gating design, this module enables controlled injection of enhanced features while cooperatively fusing multifrequency-band information, which effectively improves edge sharpness and texture fidelity in the final reconstructed image.

The structure of this article is organized as follows. Section II reviews related work on cloud removal from optical remote sensing imagery. The model design of the SAMCE-CR method in detail is introduced in Section III introduces. Section IV delineates the experimental design, encompassing experimental data, comparative methods, the experimental environment, parameter configurations, and quantitative evaluation metrics, while comprehensive experimental analyses and ablation studies on the cloud removal results are presented in Section V. Finally, Section VI draws the relevant conclusions.

II. Related Work

This section reviews the current research landscape of cloud removal techniques for remote sensing imagery, including two categories: single-modal optical image cloud removal approaches and multimodal cloud removal approaches that leverage SAR data as auxiliary information.

A. Single-Modal Optical Image Cloud Removal

The single-modal method primarily leverages the spatial context or temporal information inherent to the optical imagery itself for inferring the missing content. Early traditional methods predominantly rely on spatial inference techniques or radiative transfer physical models. Among them, methods based on spatial inference, such as Kriging interpolation and TV regularization, utilize the edge pixels of uncovered regions to achieve smooth transitions [15]; methods grounded in the physical model of radiative transfer attempt to invert cloud thickness for the restoration of true surface reflectance. However, such methods often presuppose stringent a priori conditions and demonstrate a marked propensity for producing substantial blur, spectral distortion, or excessive smoothing when processing spatially extensive, optically thick cloud regions [16].

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With the proliferation of deep learning methods, approaches predicated upon convolutional neural networks and generative adversarial networks have progressively emerged as the prevailing paradigm. Among them, CNN-based methods predominantly adopt U-Net-like encoder–decoder architectures, with an emphasis on restoring corrupted regions via multiscale feature extraction. For instance, ThiefCloud [17] employs the classic U-Net architecture as the backbone network and introduces a self-supervised learning-based cloud thickness prior along with a physical guidance feature fusion module, effectively optimizing the removal performance for thin clouds. GAN-based models focus more on enhancing the visual fidelity of reconstructed images through adversarial learning. Among them, the spatial attention generative adversarial network incorporates a spatial attention mechanism into the generator design, guiding the network to accurately localize and focus on restoring the damaged cloud regions through local-to-global weighting strategies [18]. Nevertheless, constrained by the local receptive field of standard convolutional kernels, both categories of methods encounter difficulties in establishing long-range structural dependencies when confronted with large-scale heavy occlusion, consequently yielding generated textures that frequently exhibit a deficiency in global coherence. Given the constraints of this bottleneck, models based on the attention mechanism and transformer architecture have demonstrated considerable promise in recent years [19]. Such methods directly realize accurate aggregation across regional contexts by calculating global attention weights. For instance, ACA-CRNet [20] introduces the attentive contextual attention module to the conventional attention mechanism. Through learning data-driven dynamic weights and biases, the model filters noise and irrelevant features to accurately capture effective long-range contextual information, which significantly enhances visual consistency in cloud removal. CloudRuler [21] directly adopts the transformer architecture, integrating physical rule constraints with a weighted dilated window attention mechanism, thereby effectively extracting global dependencies to achieve cloud removal in Landsat imagery.

However, single-modal methods are subject to inherent information bottlenecks, particularly under conditions of extensive cloud contamination or highly complex background textures. The recovery of authentic ground object details through spatial information inference alone remains challenging, frequently yielding structural distortions or artifacts in the reconstructed outputs.

B. SAR-Assisted Multimodal Cloud Removal

To circumvent the information bottleneck inherent to single-modal optical imagery under conditions of heavy occlusion, the introduction of SAR data, characterized by its all-weather penetration capability, has emerged as the prevailing paradigm for cross-modal assisted reconstruction [22], particularly in addressing the restoration of ground object details [23], [24]. These methods primarily learn the nonlinear mapping relationship between SAR and optical images and leverage feature fusion or image translation techniques to reconstruct authentic information obscured by clouds [25], [26].

In early investigations, scholarly attention was mainly focused on the direct mapping of SAR features into the optical domain. For example, DSen2-CR establishes a benchmark framework grounded in deep residual neural networks [27]. This method achieves effective reconstruction of thick cloud regions through the fusion of Sentinel-1 SAR imagery with cloud-contaminated Sentinel-2 optical imagery, coupled with cloud-adaptive regularization. While DSen2-CR demonstrates the feasibility of multimodal fusion, the straightforward concatenation of features proves inadequate for capturing intricate textural details. To mitigate noise transmission and texture distortion arising from the direct fusion of heterogeneous features, subsequent investigations have progressively shifted toward refined feature decoupling and multiscale interaction of deep features [28], [29]. GLF-CR proposes an SAR-enhanced global–local fusion strategy [30]. Leveraging Sentinel-1 imagery, this approach employs an SAR-guided global context interaction block to preserve overall structural consistency, while simultaneously utilizing an SAR local feature compensation block integrated with dynamic filtering to suppress speckle noise and transmit complementary textural information, thereby achieving high-precision cloud removal. In addition, to address the issue of mismatch between SAR and optical images, Align-CR designs an implicit alignment mechanism based on low-resolution SAR images, and corrects the feature mapping of multimodal and multiresolution data in the image reconstruction process, thereby attempting to compensate for the spatial misalignment between data [31]. Recent research has further elucidated intricate generative mechanisms and heterogeneous collaborative architectures. In terms of generation mechanisms, AFR-CR [32] and MT_GAN [33] have incorporated frequency-domain feature reconstruction and generative adversarial networks into cross-modal translation, while recent state-of-the-art works have further introduced diffusion models to enhance both spectral fidelity and temporal consistency. For example, RESTORE-DiT, proposed by Shu et al. [34], integrates sequence-level optical-SAR fusion into a diffusion framework, achieving highly reliable satellite image time series reconstruction even under persistent cloud cover and highly dynamic surface scenarios. In terms of architectural innovation, the hybrid architecture combining CNN and transformer has been extensively investigated to balance local detail preservation and global semantic coherence. Furthermore, cross-modal synergistic mechanisms have continued to deepen; for instance, PolNet-CR [35] systematically integrates polarimetric target decomposition features and the covariance matrix phase information of PolSAR into optical cloud removal, constructing a spatial-channel collaborative interaction network that effectively mitigates both the domain gap between optical and SAR modalities and the inherent speckle noise of PolSAR. Among them, the SAG-CR model effectively overcomes texture blurring and spectral distortion, issues commonly encountered by single-network architectures in heavy thick cloud scenarios, through the parallel coupling of convolutional layers and a Swin transformer, complemented by a cross-scale bidirectional feature transfer mechanism [28].

Although the aforementioned methods have achieved notable advancements in multimodal feature fusion and

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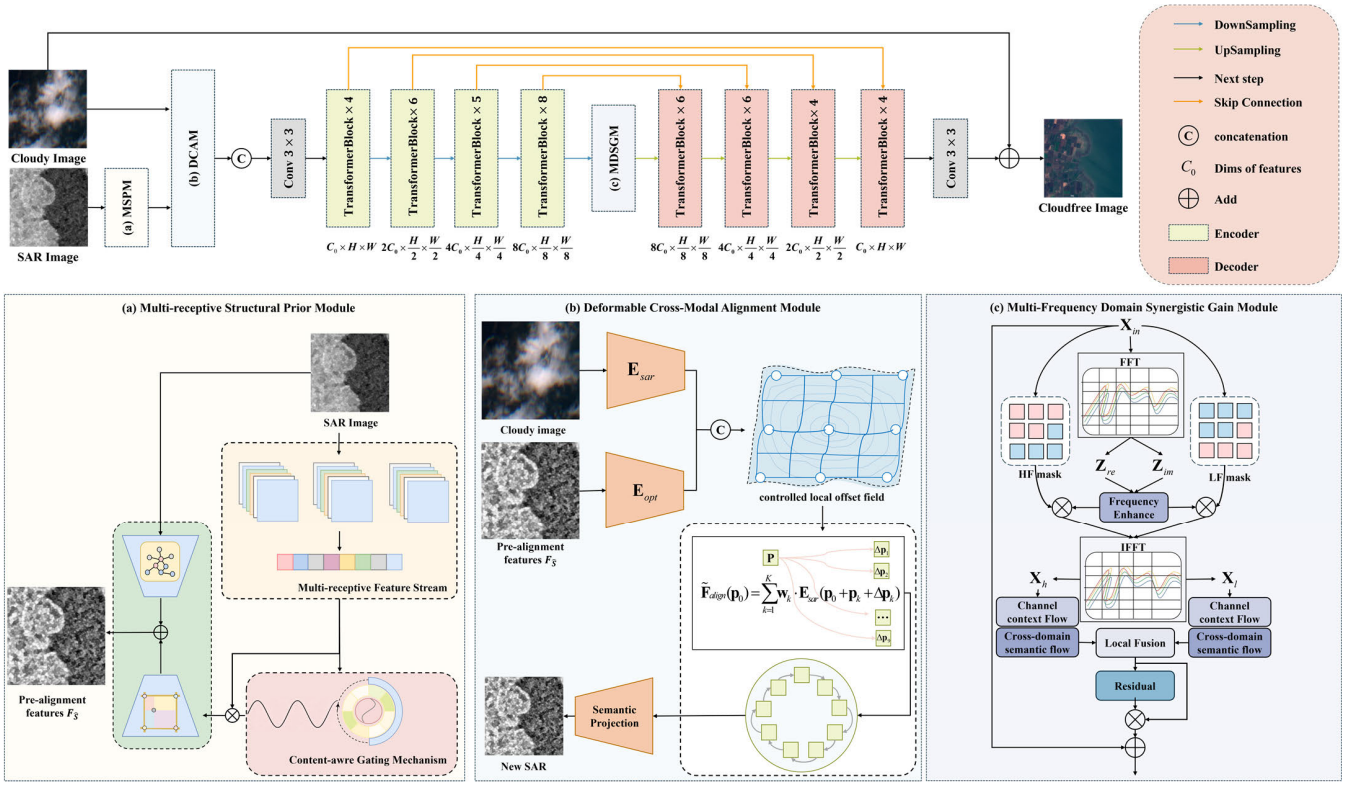


Fig. 1. SAMCE-CR frame diagram. (a) MSPM. (b) DCAM. (c) MDSGM.

deep representation learning, they remain constrained by the absence of collaborative modeling of multisource data across geometric space alignment, frequency domain component decoupling, and signal-to-noise ratio optimization, resulting in limited texture reconstruction capability in complex scenarios.

III. Methodology

The proposed method includes three components: The multireceptive structural prior module (MSPM), the deformable cross-modal alignment module (DCAM), and the multifrequency domain synergistic gain module (MDSGM). In the structure prior enhancement module, the MSPM is employed to suppress coherent speckle noise and restore spatial texture, thereby optimizing the characterization capability of auxiliary modes. In the cross-modal alignment module, the DCAM is introduced to capture pixel-level offsets and rectify SAR features to the optical reference frame, which effectively alleviates geometric mismatch. In the collaborative reconstruction module, the MDSGM is embedded to enable adaptive frequency-domain decoupling, where global structural and edge details are separately enhanced, which facilitates efficient interaction and controlled injection between the frequency and spatial domains.

The experimental procedure comprises four steps: structural prior extraction, cross-modal geometric rectification, multilayer feature encoding coupled with multifrequency domain collaborative reconstruction, and progressive decoding coupled with image generation. Fig. 1 presents the proposed framework and processing flow, which are implemented according to the following steps. Crucially, structural prior extraction must serve as a prerequisite for cross-modal alignment. This sequence is driven by the physical characteristics of the two modalities. While optical images suffer from severe structural loss due to cloud occlusion, SAR signals can penetrate clouds but are inherently corrupted by speckle noise, exhibiting both significant hierarchical feature heterogeneity and nonlinear geometric distortions. Although the cross-modal dataset images are typically coarsely co-registered, directly applying geometric correction to the raw features limits the accuracy of pixel-level fine registration. The network struggles to establish precise spatial correspondences between speckle-corrupted SAR data and heavily masked optical targets. Therefore, the prior extraction step is introduced as a necessary denoising phase. By suppressing speckle noise, it extracts clear structural features from the SAR data. These structural features provide a reliable reference, guiding the subsequent alignment module to achieve highly accurate cross-modal registration.

Structure Prior Extraction: The original noisy SAR image is first fed into a multiscale parallel network, where the spatial contours and textural details of ground objects are extracted through forward propagation. Subsequently, the spatial mask is computed via the gating unit, whereby coherent speckle noise is dynamically attenuated to refine boundary signals. Ultimately, high-quality prealignment features are generated and fed directly into the downstream alignment module, providing robust foundational structural guidance for subsequent cross-modal pixel-level alignment.

Cross-Modal Geometric Correction: First, the pure SAR image and cloud-containing optical image output from the preceding stage are jointly received, and the 2-D spatial

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displacement field for SAR features is predicted via network forward propagation with the optical data serving as a fixed reference. Then, based on the offset coordinates, local deformable sampling and mesh rearrangement are performed on the SAR features. Finally, the cross-modal fusion features, which achieve pixel-level registration, are output and subsequently fed into the encoding network.

Multilayer Feature Coding and Multifrequency Domain Cooperative Reconstruction: The registered cross-modal fusion features are sequentially fed into cascaded coding Transformer blocks, wherein multiscale high-dimensional latent representations are extracted through progressive layer-wise down-sampling. Then, FFT is performed in the deep network, which is decoupled into high- and low-frequency independent branches and utilizes multidimensional attention for collaborative reinforcement. Finally, it is transformed back to the spatial domain via the iFFT and subsequently fed into the decoding network as a high-quality deep prior.

Progressive Decoding and Image Output: First, the deep prior features from the preceding stage are sequentially fed into the decoding Transformer blocks, where they undergo dynamic fusion with same-scale skip-connection features from the encoding pathway during stepwise up-sampling. The aggregated feature tensors are then processed by a 3×3 convolutional layer to achieve spatial integration, thereby generating a detail residual map. Finally, it is directly superimposed onto the original cloud-containing optical image at the terminus of the network to yield high-quality cloud-free reconstruction results.

A. Multireceptive Structural Prior Module

To fully leverage the advantage of under-cloud structure fidelity of SAR imagery while mitigating the bottleneck in cross-modal fusion induced by speckle noise, this article proposes an MSPM module as the precursor to cross-modal fusion, which enhances the original low-resolution and speckle-corrupted SAR images, as illustrated in Fig. 2 (multiscale structural prior module).

To effectively extract clear structural priors from speckle-corrupted SAR data, the network must simultaneously capture large-scale global contours and local fine-grained edges. Therefore, the MSPM is explicitly designed with parallel multireceptive-field feature streams to extract robust structural representations across different scales. Specifically, given an input SAR image $\mathbf{F}_{sar} \in R^{2\times H\times W}$, the module ultimately outputs high-quality prealignment features $\mathbf{F}_{\widetilde{\mathbf{S}}} \in R^{2\times rH\times rW}$, where r is the up-sampling magnification, $r = 2$. First, the input undergoes parallel multiscale extraction, followed by cross-channel dimensionality reduction and fusion through concatenation and a 1×1 convolution. This generates the preliminary feature $\mathbf{F}_{fuse}$, as formulated in the following equation:

$$\mathbf{F}_{fuse} = \delta\left(\mathcal{C}_{1\times1}\left(\left[\delta\left(\mathcal{C}_{9\times9}\left(\mathbf{F}_{sar}\right)\right), \delta\left(\mathcal{C}_{5\times5}\left(\mathbf{F}_{sar}\right)\right), \delta\left(\mathcal{C}_{3\times3}\left(\mathbf{F}_{sar}\right)\right)\right]\right)\right) \tag{1}$$

where $\mathcal{C}_{k\times k}(\cdot)$ represents the convolution operation with a convolution kernel size of $k \times k$. A large-sized kernel (9×9) is used to capture global contours, and medium- and small-sized kernels (5×5, 3×3) focus on local textures and fine-grained edges, respectively. $\delta(\cdot)$ represents the PReLU activation function, which is used to introduce nonlinearity and improve the characteristic responsiveness of the model in the negative region. $[\ldots,\ldots,\ldots]$ represents a concatenation operation along the channel dimension.

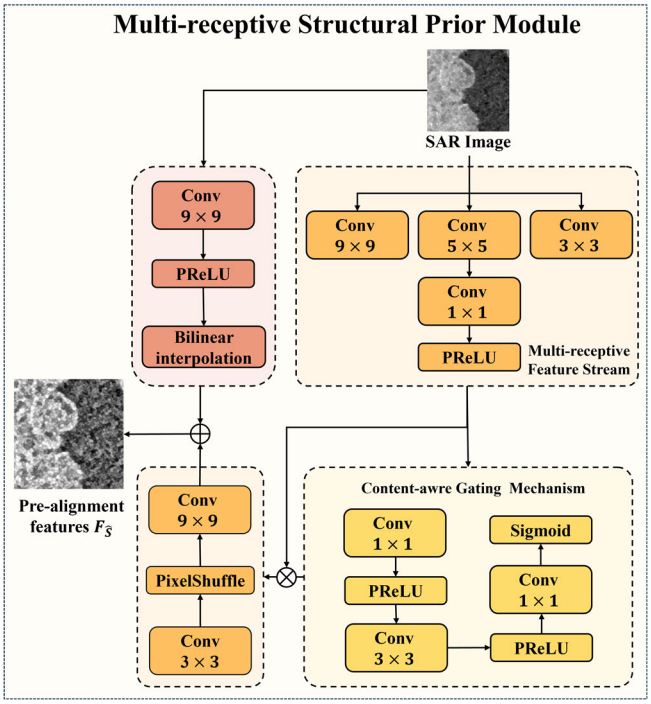


Fig. 2. Structure of MSPM.

To enable dynamic screening of texture sources, a content-aware gating mechanism is incorporated to dynamically extract contextual information from input features via the convolutional layers of the bottleneck structure, thereby adaptively inferring the spatial mask. This mask will be normalized by a Sigmoid function and multiplied in an element-wise manner with $\mathbf{F}_{fuse}$ to obtain the enhanced feature $\mathbf{F}_{enh}$

$$\mathbf{F}_{enh} = \mathbf{F}_{fuse} \odot \sigma\left(\mathcal{C}_{1\times1}\left(\delta\left(\mathcal{C}_{3\times3}\left(\delta\left(\mathcal{C}_{1\times1}\left(\mathbf{F}_{fuse}\right)\right)\right)\right)\right)\right) \tag{2}$$

where $\sigma(\cdot)$ represents the Sigmoid activation function. $\odot$ denotes element-wise multiplication.

Upon acquisition of the enhanced features, the backbone network employs pixel shuffle ($\mathcal{PS}$) to achieve artifact-free efficient up-sampling and generate high-frequency reconstructed components $\mathbf{F}_{\widetilde{S}}$

$$\mathbf{F}_{\widetilde{S}} = \mathcal{C}_{9\times9}\left(\mathcal{PS}\left(\mathcal{C}_{3\times3}\left(\mathbf{F}_{enh}\right)\right)\right). \tag{3}$$

Simultaneously, to mitigate the gradient divergence issue inherent in deep networks during the initial training phase and to furnish a stable low-frequency prior, a global residual pathway is introduced in parallel within this framework to generate low-frequency approximate components $\mathbf{R}(\mathbf{S})$ via bilinear interpolation ($\mathcal{UP}$)

$$\mathbf{R}\left(\mathbf{S}\right) = \mathcal{UP}\left(\delta\left(\mathcal{C}_{9\times9}\left(\mathbf{F}_{sar}\right)\right)\right). \tag{4}$$

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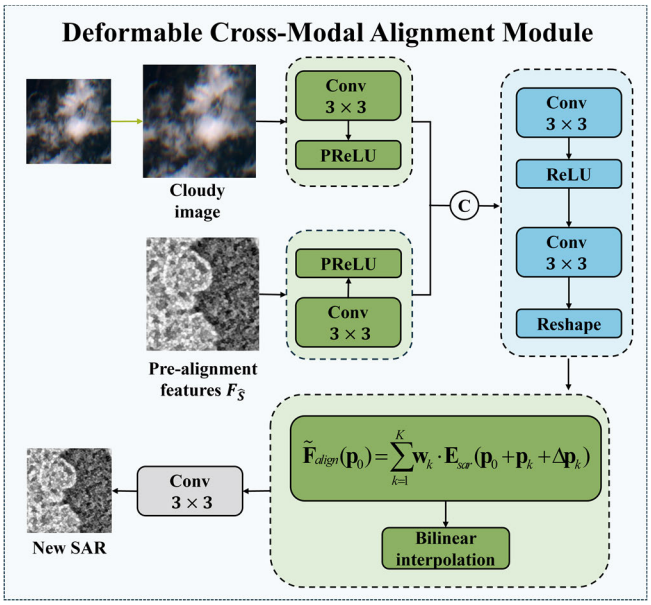


Fig. 3. Structure of DCAM.

Ultimately, the output $\mathbf{F}_{\widehat{S}}$ of the MSPM is superimposed by high-frequency details and low-frequency profiles

$$\mathbf{F}_{\widehat{S}} = \mathbf{F}_{\widetilde{S}} + \mathbf{R}(\mathbf{S}). \tag{5}$$

The output features $\mathbf{F}_{\widetilde{S}}$ contain high-quality structural information, which are used as the input to the subsequent DCAM, effectively reducing the offset estimation instability problem caused by low-quality signals.

B. Deformable Cross-Modal Alignment Module

To address geometric discrepancies, variations in viewing angles, and ground object scattering characteristics between SAR and optical in phase and passive imaging, as well as cross-scale misalignments arising from inconsistent native resolutions, this article proposes the DCAM to perform deformable alignment of SAR features, which are enhanced by the MSPM, within the reference framework of optical features, as illustrated in Fig. 3 (DCAM). To compute geometric correspondence within a unified high-resolution feature space, the optical reference feature $\mathbf{F}_{opt} \in R^{4\times H\times W}$ is first upsampled via adaptive average pooling to match the spatial dimensions of the enhanced SAR features $\mathbf{F}_{\widehat{\mathbf{S}}} \in R^{2\times rH\times rW}$. The upsampled optical feature is denoted as $\mathbf{F}_{\widehat{opt}} \in R^{4\times rH\times rW}$. Subsequently, DCAM performs nonlinear feature embedding on $\mathbf{F}_{\widetilde{S}}$ and $\mathbf{F}_{\widehat{opt}}$ to extract their respective shallow semantic features, $\mathbf{E}_{sar}$ and $\mathbf{E}_{opt}$, while maintaining their original channel dimensions

$$\mathbf{E}_{sar} = \delta\left(\mathcal{C}_{3\times3}\left(\mathbf{F}_{\widehat{S}}\right)\right) \tag{6}$$

$$\mathbf{E}_{opt} = \delta\left(\mathcal{C}_{3\times3}\left(\mathbf{F}_{\widehat{opt}}\right)\right). \tag{7}$$

Subsequently, the two are concatenated along the channel dimension to produce a joint tensor, following which a controlled local offset field $\mathbf{O} \in \mathrm{R}^{2K^2\times H\times W}$ is predicted via a lightweight convolutional network:

$$\mathbf{O} = \mathcal{C}_{3\times3}\left(\ell\left(\mathcal{C}_{3\times3}\left(\left[\mathbf{E}_{sar}, \mathbf{E}_{opt}\right]\right)\right)\right) \tag{8}$$

where $\ell(\cdot)$ is the RELU activation function. The generated offset field $\mathbf{O}$ encodes the spatial displacement vector required for each sampling point on $\mathbf{F}_{\widehat{S}}$ when considering $\mathbf{F}_{opt}$ as a geometric reference. Here, K denotes the spatial kernel size of the deformable convolution, yielding a total of K^2 sampling points, and $2K^2$ represents the total coordinate offsets for these sampling points along the spatial x- and y-directions.

Based on the predicted offset field $\mathbf{O}$, a deformable convolution is performed on the input features $\mathbf{E}_{sar}$ to achieve pixel-level geometric correction. Unlike standard convolution, which operates on a fixed regular grid, this operation dynamically adjusts sampling positions through learned offsets to reconstruct SAR features within the optical reference frame. Reshape reorganizes the offset tensors $2\times9\times H\times W$ into nine sampling points corresponding to each spatial position, with each sampling point comprising x- and y-coordinate offsets to facilitate subsequent sampling operations. The calculation formula for the aligned feature $\widetilde{\mathbf{F}}_{align}$ at spatial position $\mathbf{p}_0$ is given by

$$\widetilde{\mathbf{F}}_{align}(\mathbf{p}_0) = \sum_{k=1}^{K} \mathbf{w}_k \cdot \mathbf{E}_{sar}(\mathbf{p}_0 + \mathbf{p}_k + \Delta\mathbf{p}_k) \tag{9}$$

where $\mathbf{w}_k$ is the weight of the convolution kernel at the kth position. $\mathbf{p}_k$ is the predefined offset on the regular grid. $\Delta\mathbf{p}_k$ is the learnable position offset at the index from the offset field $\mathbf{O}$. In practice, the eigenvalues of the continuous space are obtained via bilinear interpolation for noninteger sampling points. Finally, the local information is reaggregated according to the updated spatial neighborhood relationships to suppress interpolation artifacts, and the features are smoothly projected into the semantic space compatible with subsequent fusion modules via 3×3 convolution, yielding the final steady-state aligned SAR features $\mathbf{F}_{align} = \mathcal{C}_{3\times3}(\widetilde{\mathbf{F}}_{align})$. Finally, the aligned SAR features are downsampled via interpolation back to the original $2\times H\times W$ scale. They are then concatenated with the original optical image $\mathbf{F}_{opt}$ to form a $6\times H\times W$ joint tensor, which is officially fed into the network for cross-modal fusion.

C. Multifrequency Domain Synergistic Gain Module

To fully exploit the complementary advantages of SAR and optical imagery and effectively mitigate cloud and fog interference, the aligned SAR features $\mathbf{F}_{align}$ generated from the preceding stage are concatenated with the optical reference features $\mathbf{F}_{opt}$ along the channel dimension. Subsequently, the joint features undergo fusion through a 3×3 convolution operation, followed by deep feature extraction via the Cascaded Encoders, yielding deep fusion features $\mathbf{X}_{in} \in \mathrm{R}^{C\times H\times W}$ enriched with semantic information. Existing spatial and channel attention mechanisms primarily reweight features based on spatial locations or channel dependencies. However, thick clouds and ground objects often overlap within the same spatial regions, making it difficult for standard attention mechanisms to effectively separate these coupled signals in the spatial domain. To address this limitation, the MDSGM is proposed in this study, as depicted in Fig. 4. The fundamental distinction of MDSGM lies in decoupling features in the frequency domain. Physically, cloud degradation manifests

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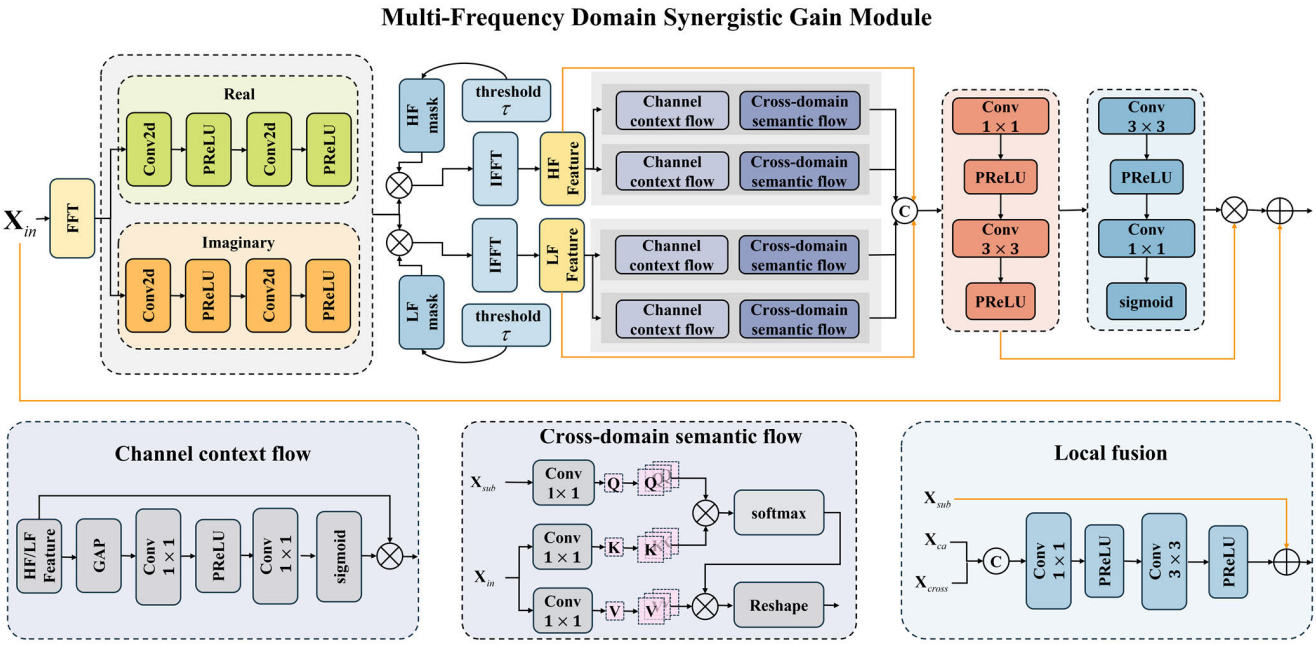


Fig. 4. Structure of MDSGM.

as low-frequency interference, while ground structural details correspond to high-frequency components. Through adaptive spectrum segmentation, dual-stream feature interaction, and gated injection, MDSGM explicitly suppresses low-frequency cloud responses and enhances high-frequency edges, achieving independent enhancement and collaborative reconstruction of sub-bands. The specific process is as follows.

To accommodate variations in cloud cover distributions, the model adaptively delineates the boundary between high- and low-frequency, corresponding to the threshold component illustrated in Fig. 4. First, the control vector is predicted from $\mathbf{X}_{in}$ via the global adaptive average pooling (GAP) followed by a 1×1 convolution, from which the high- and low-frequency truncation parameters are extracted

$$\tau = 0.5 \cdot \sigma\left(\mathcal{C}_{1\times1}\left(\mathrm{GAP}\left(\mathbf{X}_{in}\right)\right)\right) \tag{10}$$

where the coefficient 0.5 is applied to strictly constrain the predicted truncation radii τ within the valid physical spectrum of $[0, 0.5]$ up to the Nyquist limit [36]. Further discussions on the physical interpretability of these learned thresholds are detailed in Section V-E5. Subsequently, these truncation radii are applied to the 2-D frequency radius map $\mathbf{R}$, which is normalized to the $[-0.5, 0.5]$ range, to generate the high-frequency binary mask $\mathbf{M}_h = \mathbb{I}(\mathbf{R} > \tau)$ and the low-frequency binary mask $\mathbf{M}_l = \mathbb{I}(\mathbf{R} \leq \tau)$. $\mathbb{I}(\cdot)$ is the standard indicator function (returning 1 if the condition is met, and 0 otherwise). This ensures an accurate and adaptive segmentation of the high- and low-frequency regions.

Subsequently, the features are transformed into the frequency domain via the FFT to obtain $\mathbf{Z} = \mathcal{F}(\mathbf{X}_{in})$. Subsequently, nonlinear convolution enhancement is applied to the real component $\mathbf{Z}_{re}$ and imaginary component $\mathbf{Z}_{im}$, respectively. The inverse transformation ($\mathcal{F}^{-1}$) is then performed in conjunction with the generated mask $\mathbf{M}_{sub} \in \{\mathbf{M}_h, \mathbf{M}_l\}$ to decompose independent high- and low-frequency spatial sub-band features $\mathbf{X}_{sub} \in \{\mathbf{X}_h, \mathbf{X}_l\}$

$$\mathbf{X}_{sub} = \mathcal{F}^{-1}\left(\left(\Phi\left(\mathbf{Z}_{re}\right) + i\Phi\left(\mathbf{Z}_{im}\right)\right) \odot \mathbf{M}_{sub}\right) \tag{11}$$

where $\Phi(\cdot)$ denotes a nonlinear feature extraction block comprising two stacked 3×3 convolutional layers with the PReLU activation function.

Given that FFT may result in partial spatial semantic loss, dual-stream features are computed in parallel for each frequency sub-band $\mathbf{X}_{sub}$. The channel context stream is designed to recalibrate channel weights by leveraging global information, thereby emphasizing salient frequency responses

$$\mathbf{X}_{ca} = \sigma\left(\mathcal{C}_{1\times1} \cdot \left(\delta\left(\mathcal{C}_{1\times1}\left(\mathrm{GAP}\left(\mathbf{X}_{sub}\right)\right)\right)\right) \odot \mathbf{X}_{sub}\right). \tag{12}$$

The cross-domain semantic stream leverages the original input $\mathbf{X}_{in}$ to guide the frequency sub-band in recovering the original spatial details. Through the corresponding linear mapping weight matrices $\mathbf{W}_Q$, $\mathbf{W}_K$, and $\mathbf{W}_V$, the current frequency sub-band $\mathbf{X}_{sub}$ is flattened and linearly projected to form Query $\mathbf{Q} = \mathbf{X}_{sub}\mathbf{W}_Q$, while the original feature $\mathbf{X}_{in}$ is projected to form the Key $\mathbf{K} = \mathbf{X}_{in}\mathbf{W}_K$ and Value $\mathbf{V} = \mathbf{X}_{in}\mathbf{W}_V$. The scaled dot-product attention is computed

$$\mathbf{X}_{cross} = \mathrm{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \tag{13}$$

where d_k is the scaling factor. The attention output is restored to the spatial feature map by reshaping to facilitate subsequent spatial processing.

Subsequently, to integrate contextual and cross-domain semantic information, the aforementioned dual-stream features are concatenated and fused, with structurally enhanced steady-state sub-band features $\widetilde{\mathbf{X}}_{sub}$ subsequently output via residual connection

$$\widetilde{\mathbf{X}}_{sub} = \mathbf{X}_{sub} + \delta\left(\mathcal{C}_{3\times3}\left(\delta\left(\mathcal{C}_{1\times1}\left(\left[\mathbf{X}_{ca}, \mathbf{X}_{cross}\right]\right)\right)\right)\right). \tag{14}$$

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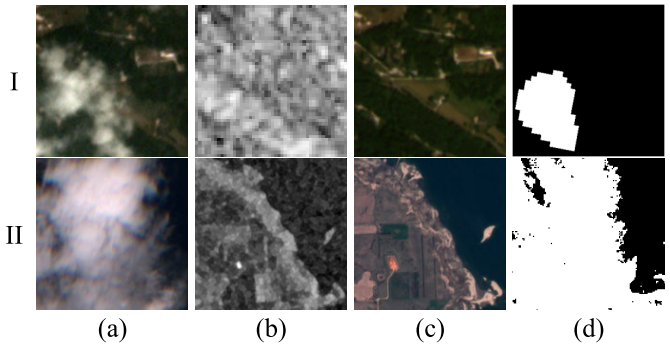


Fig. 5. Example image pairs from the two remote sensing datasets used in the experiments. (I) The M3R-CR dataset and (II) the LuojiaSET-OSFCR dataset. For each dataset, the columns from left to right display. (a) Cloudy optical image. (b) Paired SAR image. (c) Cloud-free reference image. (d) Cloud mask.

Upon completion of cross-frequency mutual conductance optimization, the modulated high-frequency features $\tilde{\mathbf{X}}_h$ and low-frequency features $\tilde{\mathbf{X}}_l$ are subsequently reintegrated into the backbone network. To prevent excessive feature injection from disrupting the original representation distribution, this article employs a spatially adaptive gating strategy. The final output feature $\mathbf{X}_{out}$ is defined as

$$\mathbf{X}_{out} = \mathbf{X}_{in} + \gamma \cdot \mathcal{G}(\cdot) \odot \delta\left(\mathcal{C}_{3\times3}\left(\delta\left(\mathcal{C}_{1\times1}\left(\left[\widetilde{\mathbf{X}}_h, \widetilde{\mathbf{X}}_l\right]\right)\right)\right)\right) \quad (15)$$

where γ is a learnable scaling factor used to adjust the injection intensity. $\mathcal{G}(\cdot)$ is a lightweight convolutional network employed to generate pixel-level gated maps, which are utilized to adaptively determine whether restoration features should be injected into cloud-covered regions while preserving clear background areas intact.

IV. Experiments Design

A. Experimental Data

1) M3R-CR Dataset [31]: This dataset comprises 63 000 sets of multimodal remote sensing imagery, each consisting of strict spatial registration of cloudy PlanetScope multispectral imagery, cloud-free PlanetScope multispectral imagery, and Sentinel-1 dual-polarization (VV, VH) SAR imagery. Among them, the PlanetScope imagery has a spatial resolution of 3 m and comprises four spectral bands: blue, green, red, and near-infrared; the Sentinel-1 SAR imagery has a spatial resolution of 10 m. Regarding temporal proximity, the average temporal gap between paired cloudy and cloud-free optical images is 2.8 days, compared to 2.2 days between cloudy optical images and SAR data.

2) LuojiaSET-OSFCR Dataset [37]: This dataset comprises 20 000 sets of spatiotemporally registered Sentinel-2 optical imagery and Sentinel-1 SAR imagery, encompassing ten nonoverlapping regions globally. This dataset provides pixel-level semantic annotations encompassing six representative land-cover categories, including water bodies and forest land, thereby enabling direct evaluation of both visual and semantic restoration quality within complex scenarios.

To better illustrate the characteristics and diversity of the data used in the experiments, Fig. 5 visualizes representative example pairs from both the M3R-CR and LuojiaSET-OSFCR datasets, including the cloudy optical images, paired SAR images, cloud-free reference images, and extracted cloud masks.

B. Methods of Comparison

To demonstrate the effectiveness of the proposed SAMCE-CR method, this article conducts a comparative analysis against eight state-of-the-art de-clouding methods. In terms of CNN and attention mechanism architectures, DSen2-CR [27], which combines cloud adaptive regularization loss, and ACA-CR [20], which introduces a contextual attention mechanism, are selected. In terms of multimodal interaction and feature alignment, this study selects GLF-CR [30], which leverages SAR to guide global–local feature interaction; Align-CR [31], which achieves heterogeneous feature alignment through deformable convolution. In terms of advanced architectures and physical priors, this article compares ECRformer [38] and AFR-CR [32], two SAR-assisted transformer networks that focus on effective multimodal feature interaction and adaptive frequency-domain feature reconstruction, respectively. CloudRuler [21], another transformer-based approach coupling a physical model with sparse attention; and ThiefCloud [17], a U-Net-based architecture that combines a self-supervised cloud prior with physically guided fusion.

C. Experimental Environment and Parameter Settings

The experimental environment was configured as follows: the operating system was Windows 10; the processor was an Intel[1] Core[2] i9-13900K; the experimental environment consisted of Python 3.8 as the programming language, PyTorch 2.0.0 as the deep learning framework, and CUDA 11.8. The graphics processing unit (GPU) was an NVIDIA RTX A5000 with 16 GB of video memory. The proposed SAMCE-CR method is implemented based on the publicly available PyTorch framework and trained in an end-to-end fashion. To enhance the robustness of the model against outliers and improve texture restoration in cloud-occluded regions, this study employs Charbonnier loss [39] for training, which exhibits superior outlier handling capability while imposing additional constraints on cloud-covered regions. Image patches of 160×160 pixels were randomly cropped from the training samples during the training phase. The batch size was specified as 4, and the maximum number of training iterations was set to 30. The Adam optimizer was adopted, with the initial learning rate set to 10^{-4} and a 50% decay applied every 5 epochs.

D. Quantitative Evaluation Indicators

This experiment employs six widely used metrics to evaluate cloud removal performance: the peak signal-to-noise ratio (PSNR) [40] and mean absolute error (MAE) [41] are utilized to quantify pixel-level statistical errors; structure similarity index measure (SSIM) [42] and spectral angle mapper (SAM) [43] evaluate the fidelity of spatial topology structure and

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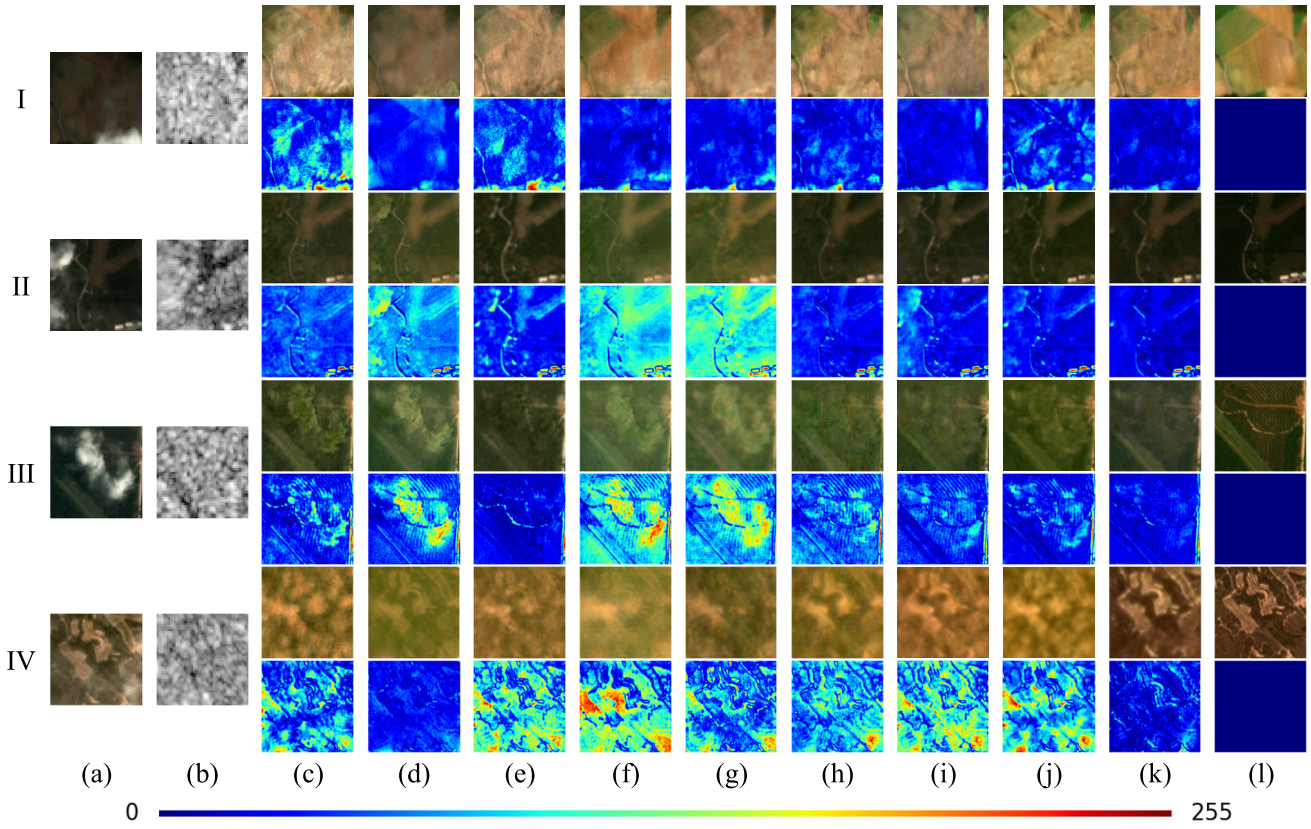


Fig. 6. Qualitative comparison on the M3R-CR dataset across four typical scenarios. For each scenario, the predicted cloud-free images are displayed above their corresponding residual maps. (a) Cloudy image. (b) SAR image. (c) DSen2-CR. (d) GLF-CR. (e) ACA-CR. (f) Align-CR. (g) ThiefCloud. (h) CloudRuler. (i) AFR-CR. (j) ECRformer. (k) SAMCE-CR. (l) Cloud-free.

spectral information, separately; and learned perceptual image patch similarity (LPIPS) [44] and Fréchet inception distance (FID) [45] measure the realism of the generated imagery from the local texture perception and global semantic distribution from the depth features. Overall, the higher the PSNR and SSIM values, the lower the MAE, SAM, LPIPS, and FID values, which indicates better reconstruction quality.

V. Experimental Results and Analysis

A. Qualitative Evaluation

Figs. 6 and 7 present the qualitative performance of our SAMCE-CR and eight comparative methods for cloud removal on the M3R-CR and LuojiaSET-OSFCR datasets, respectively. For a comprehensive evaluation, four representative test scenarios were selected from each dataset to facilitate comparative analysis. To more intuitively reflect the recovery degree of local details, for each test scenario, the figure presents not only the cloud-free image reconstructed by each method but also the residual map between the reconstructed result and the ground truth image. In the residual maps, darker blue hues correspond to smaller residuals, whereas darker red hues correspond to larger residuals.

1) Visual Contrast Analysis on the M3R-CR Dataset: As illustrated in Fig. 6, Scenario I depicts a multiseasonal farmland splicing area, where the staggered distribution of fields induces substantial spectral discrepancies among ground object features. This analysis primarily examines the model's capability to restore the edges and spectral information of ground objects. Observation and comparison results indicate that DSen2-CR, ACA-CR, and CloudRuler exhibit limited processing capability in thick cloud regions, resulting in blurred textures beneath cloud cover. The corresponding residual maps display prominent yellow-green high reconstruction error patches, which indicate low reconstruction accuracy. Although ECRformer effectively removes cloud noise, the resulting images exhibit noticeable shadows and deviate markedly from the ground truth. Meanwhile, the reconstruction results of both GLF-CR and ThiefCloud exhibit conspicuous artifacts and high-frequency noise, which compromise the texture continuity of the agricultural landscape. While Align-CR and AFR-CR achieve favorable results in areas with thin cloud cover, both methods exhibit noticeable texture loss under thick cloud occlusion. Specifically, although Align-CR produces visually smooth imagery in agricultural areas, its corresponding residual map reveals larger errors along field boundaries, indicating an over-smoothing tendency that degrades structural geometric integrity. In contrast, SAMCE-CR achieves the best visual quality. Leveraging the SAR prior, the proposed method accurately restores the spectral characteristics of diverse plots while clearly reconstructing ridge boundaries. The residual map exhibits minimal, evenly distributed errors, validating the model's capacity for high-fidelity structural reconstruction.

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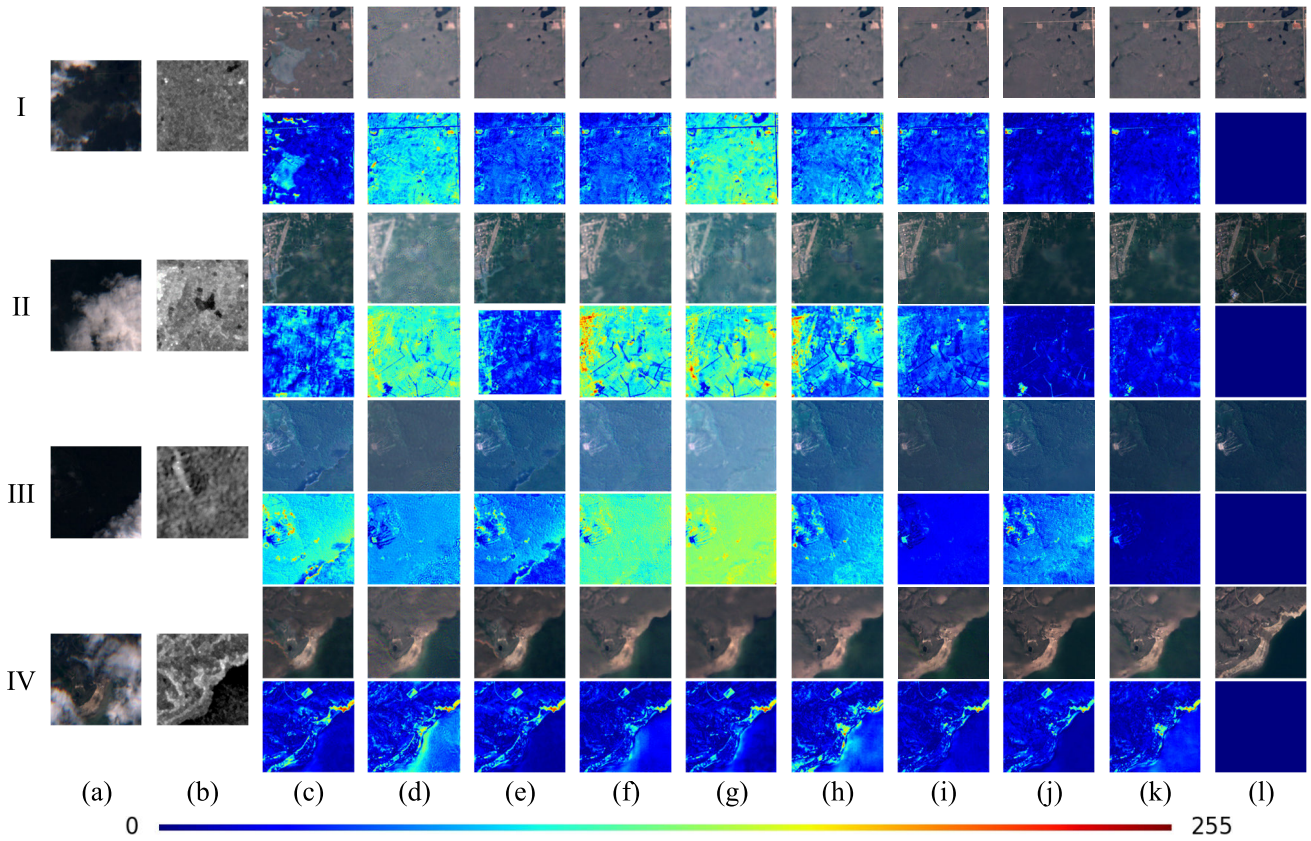


Fig. 7. Qualitative comparison on the LuojiaSET-OSFCR dataset across four typical scenarios. For each scenario, the predicted cloud-free images are displayed above their corresponding residual maps. (a) Cloudy image. (b) SAR image. (c) DSen2-CR. (d) GLF-CR. (e) ACA-CR. (f) Align-CR. (g) ThiefCloud. (h) CloudRuler. (i) AFR-CR. (j) ECRformer. (k) SAMCE-CR. (l) Cloud-free.

Scenario II represents a vegetation-interleaved road network area where prominent linear road features traverse complex backgrounds, testing the model's resilience in maintaining the continuity and integrity of slender roads. Experimental results show that the GLF-CR method generates granular artifacts during cloud removal, which consequently impairs the textural smoothness of vegetation regions. Align-CR fails to effectively utilize spatial contextual information, leading to the blurring or even fragmentation of linear features within occluded road segments. Although ACA-CR, CloudRuler, and ECRformer restore the general outlines, a hazy appearance induced by cloud and haze persists at junctions where the road network and vegetation form strong contrast. In contrast, the proposed method exhibits excellent structural remodeling capability. The residual maps reveal a low error response in high-frequency regions, such as road edges. This substantiates that the proposed method not only fully restores the geometric structure of winding road networks in the visual domain, thereby avoiding the fragmentation of linear features, but also effectively mitigates blurring effects in mixed regions while preserving the sharpness of ground object boundaries.

Scenario III represents a heterogeneous stand boundary area characterized by abrupt surface texture transitions along the boundary line, thereby challenging the model's reconstruction accuracy of structural boundaries under cloud occlusion conditions. Visual comparisons reveal that, it is observed that Align-CR produces prominent yellow patch-like artifacts in the cloud removal regions, leading to severe spectral distortion and texture degradation. Consequently, the corresponding regions in its residual map exhibit marked red errors. DSen2-CR, GLF-CR, and ThiefCloud exhibit limited efficacy in capturing texture transitions, leading to ambiguous boundaries between woodland and bare land and indistinct edge features. Although AFR-CR and ECRformer are capable of cloud removal, they exhibit limitations in restoring high-frequency details, resulting in vegetation areas that appear overly smooth and lack realistic texture fidelity. In contrast, the proposed method benefits from the effective constraint on edge information, which not only accurately restores the clear and sharp boundaries of ground objects but also faithfully reconstructs the regularly arranged high-frequency texture features in woodland areas. The homogeneous distribution of dark blue across the residual map further corroborates that the proposed method achieves both exceptional spectral fidelity and structural precision in regions characterized by pronounced textural discontinuities.

Scenario IV represents a complex gully-textured terrain characterized by densely distributed, winding dendritic channels. This high-frequency spatial texture serves to evaluate the model's capacity for capturing tiny topographic details while mitigating smoothing effects. As evidenced by the experimental results, AFR-CR achieves cloud removal at the cost of substantial color deviation and excessive texture smoothing, leading to the near-total disappearance of gully details. DSen2-CR and Align-CR exhibit instability in handling

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complex backgrounds beneath cloud-covered regions. The former generates obvious blurred bands at junction regions, whereas the latter introduces extraneous noise that compromises surface spatial consistency. Although CloudRuler and GLF-CR successfully restore the macroscopic contours, they exhibit varying degrees of information loss and artifact interference at the microscopic scale, thereby failing to reproduce the original surface roughness. In contrast, the proposed method leverages the accurate capture of multiscale features, which achieves not only a visually perfect reconstruction of dense dendritic textures while avoiding fine-structure fractures and blurring, but also maintains a high degree of spectral consistency with the actual surface.

2) Visual Contrast Analysis on the LuojiaSET-OSFCR Dataset: As illustrated in Fig. 7, Scenario I comprises a mixed area of bare land and water bodies, characterized by the coexistence of high-reflectance bare land and low-reflectance water bodies, which readily induces spectral distortion in cloud removal models. Fig. 7 reveals a marked spectral displacement between GLF-CR and ThiefCloud, leading to considerable discrepancies in chromaticity between the generated images and real ground objects. DSen2-CR exhibits instability when processing complex features at land-water boundaries, introducing blocky artifacts and noise that roughen the originally smooth surface areas. ACA-CR and CloudRuler fail to achieve complete cloud removal in thick cloud regions, resulting in residual haze that impairs the visibility of ground objects. In contrast, ECRformer and the proposed method effectively circumvent hue smoothing issues during cloud removal, preserve the highlighted signatures of bare ground, and accurately reconstruct the low-reflectance characteristics of water bodies without introducing perceptible artifacts. The residual maps demonstrate that the error response of the cloud-removed regions is maintained at a low level, objectively demonstrating the reliability of the proposed method for handling regions with mixed surface contrasts.

Scenario II is a mixed area of artificial vegetation in the suburbs, intertwined with artificial buildings with regular geometric characteristics and natural vegetation with irregular textures. The qualitative results demonstrate that existing methods struggle to reconcile these two disparate characteristics concurrently. Both AFR-CR and ACA-CR exhibit pronounced deviations in radiance brightness, with the overall image contrast being substantially reduced. Consequently, the demarcation between roads and buildings becomes indistinguishable against the subdued backgrounds. GLF-CR, Align-CR, and ThiefCloud introduce over-smoothing effects when processing high-frequency geometric structures, resulting in blurred building edges and loss of geometric sharpness of artificial features. DSen2-CR, ACA-CR, and CloudRuler generate unnatural artifacts and noise at complex texture boundaries, which compromise the spatial continuity of the suburban regions. In contrast, the proposed method demonstrates superior structural reconstruction capability. It not only accurately restores the distinct geometric contours of the building complex and road network but also effectively preserves the inherent spectral characteristics of the surrounding vegetation, thereby circumventing structural ambiguities and edge artifacts. In the residual map, even within the mixed boundary zone exhibiting the most intricate texture, the proposed method maintains a low error response, which demonstrates its robustness in handling complex mixed ground objects.

Scenario III is situated within a complex mountainous terrain characterized by pronounced topographic relief and dense forest vegetation. Combined with visual comparison and residual map analysis, it is obvious that GLF-CR, CloudRuler, and ECRformer suffer from obvious brightness suppression in this scene. Although these methods do not generate extensive red and yellow high-error regions, their outputs differ substantially from Cloud-free. While DSen2-CR and ACA-CR recover approximate hues, the conspicuous red-yellow high-error patches in their residual maps reveal severe loss of texture information within cloud-covered regions. Align-CR and ThiefCloud struggle to address illumination variations induced by undulating terrain. They not only fail to reconstruct the distinct ridgeline but also introduce speckle noise, which compromises the inherent spatial continuity of mountainous areas. In contrast, the proposed method successfully decouples cloud occlusion from topographic shadows and distinctly reconstructs undulating mountain ridges and fine-scale vegetation textures, thereby demonstrating its superior capability in high-frequency topographic detail reconstruction.

Scenario IV is a coastal water area characterized by pronounced spectral disparities between water bodies and land. This scenario primarily evaluates the model's capacity for spectral consistency restoration within homogeneous water regions. DSen2-CR, ACA-CR, and Align-CR fail to effectively constrain pixel diffusion at the land-water junction, resulting in shoreline blurring. This is quantitatively confirmed by the corresponding residual map, where the shoreline location presents a significant red-yellow high-error band, indicating that these models struggle to fit the high-frequency gradient variations in this region. GLF-CR and CloudRuler are primarily constrained by spectral distortion in the water. Among them, CloudRuler introduces dense speckle-like artifacts in the water, whereas GLF-CR excessively dims the water body brightness, leading to a complete loss of detail in shallow sea areas. In contrast, while heavily smoothed methods like AFR-CR may produce visually clean results in homogeneous water regions by sacrificing high-frequency details, SAMCE-CR strives to maintain overall structural fidelity by rigorously leveraging SAR priors. However, it is worth noting that in certain highly dynamic coastal boundaries, the inherent speckle noise and backscatter fluctuations of SAR imagery over water bodies can occasionally introduce localized high-frequency interference, leading to slight spectral artifacts at the land-water junction. Nevertheless, across diverse global landscapes, the proposed method achieves the most favorable balance between geometric structural preservation and spectral recovery.

B. Quantitative Evaluation

To further objectively evaluate the reconstruction performance of various cloud removal methods and overcome the subjectivity inherent in purely visual interpretation, this section

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TABLE I
QUANTITATIVE EVALUATION INDICATORS OF THE M3R-CR DATASET

Method	PSNR↑	SSIM↑	SAM↓	MAE↓	FID↓	LPIPS↓
DSen2-CR (2020)	30.0888	0.8962	6.1990	0.0260	48.7043	0.1177
GLF-CR (2022)	30.1114	0.8989	6.1776	0.0261	55.2883	0.1405
ACA-CR (2024)	30.0934	0.8955	6.4218	0.0265	41.5225	0.1059
Align-CR (2024)	29.9415	0.8967	5.8116	0.0275	68.3911	0.1583
ThiefCloud (2025)	30.1426	0.8955	5.8476	0.0261	82.0588	0.2024
CloudRuler (2025)	30.8129	0.9106	5.6720	0.0249	61.3700	0.1450
AFR-CR (2026)	31.0586	0.9031	6.0496	0.0244	41.3449	0.1052
ECRformer (2026)	30.4444	0.9074	5.8951	0.0260	54.5999	0.1183
SAMCE-CR	**31.6229**	**0.9128**	**5.4370**	**0.0229**	**35.4971**	**0.0901**

TABLE II
QUANTITATIVE EVALUATION INDICATORS OF THE LUOJIASET-OSFCR DATASET

Method	PSNR↑	SSIM↑	SAM↓	MAE↓	FID↓	LPIPS↓
DSen2-CR (2020)	32.1701	0.9119	3.0657	0.0191	73.7647	0.1499
GLF-CR (2022)	34.2967	0.9183	2.5553	0.0144	42.1871	0.1611
ACA-CR (2024)	33.2008	0.9218	2.7365	0.0170	64.6018	0.0920
Align-CR (2024)	35.3003	0.9313	2.2290	0.0127	45.7030	0.0979
ThiefCloud (2025)	33.7010	0.9252	2.6925	0.0156	68.7843	0.2126
CloudRuler (2025)	35.6582	0.9389	2.1154	0.0122	42.4951	0.1587
AFR-CR (2026)	36.2185	0.9418	1.9993	0.0114	35.5703	0.0638
ECRformer (2026)	36.0958	**0.9453**	1.9826	0.0116	38.8241	0.1720
SAMCE-CR	**36.4149**	0.9433	**1.9510**	**0.0111**	**33.3958**	**0.0572**

adopts multidimensional quantitative indicators to comprehensively compare the experimental results. Tables I and II show the quantitative evaluation results of various de-clouding methods on the M3R-CR dataset and the LuojiaSET-OSFCR dataset, respectively. To facilitate performance comparison, the optimal results are presented in boldface, with the suboptimal results indicated by underlining.

1) Analysis of Quantitative Evaluation Results on the M3R-CR Dataset: The quantitative evaluation presented in Table I demonstrates that the proposed method achieves superior performance across six evaluation metrics on the M3R-CR dataset, effectively alleviating the tradeoff dilemma inherent in existing methods. Regarding fundamental reconstruction fidelity, although AFR-CR maintains a high signal-to-noise ratio, the performance of the proposed method is superior, achieving a PSNR of 31.6229 dB, which represents an improvement of 0.5643 dB over AFR-CR. In addition, the minimal MAE value of 0.0229 provides further objective evidence of the efficacy of the proposed method in suppressing pixel-level errors. In the spectral fidelity evaluation, compared with methods such as ACA-CR and Align-CR, the SAM value of the proposed method is optimized to 5.4370, representing a reduction of 0.2350 relative to the suboptimal value of 5.6720 achieved by the CloudRuler method, thereby ensuring high spectral consistency between the cloud-removed image and the real ground objects. More significant differences are reflected in the perceived quality indicators. The LPIPS value of the ThiefCloud method is 0.2024, indicating that the images generated by this method exhibit certain deviations from the reference ground truth in terms of texture details. In contrast, the proposed method achieves a lower FID value of 35.4971 and an LPIPS value of 0.0901. This outcome demonstrates that images generated by SAMCE-CR exhibit enhanced fidelity to real-world objects at the perceptual level, achieving an effective equilibrium between objective metrics and subjective visual quality.

2) Analysis of Quantitative Evaluation Results on the LuojiaSET-OSFCR Dataset: Table II presents the quantitative evaluation results obtained by each method on the LuojiaSET-OSFCR dataset. The proposed method achieves optimal values across five evaluation indices and shows robust comprehensive performance. In terms of image reconstruction accuracy, the proposed method achieves a PSNR value of 36.4149 dB and an SSIM value of 0.9433, with the MAE maintained at a minimum of 0.0111. These quantitative metrics demonstrate the superior performance of the proposed method relative to the comparative methods, indicating enhanced signal recovery capability and effective error suppression. In the evaluation of spectral dimension, the SAM value of the proposed method is 1.9510, confirming that the proposed method maintains satisfactory spectral consistency throughout the fusion process. In addition, the proposed method attains an FID value of 33.3958 and an LPIPS value of 0.0572, both representing optimal performance. These quantitative results indicate that the generated images exhibit superior visual realism in detail preservation, aligning more closely with human visual perception. In contrast, ThiefCloud exhibits a pronounced disparity in perceptual metrics, with an LPIPS value of 0.2126. This

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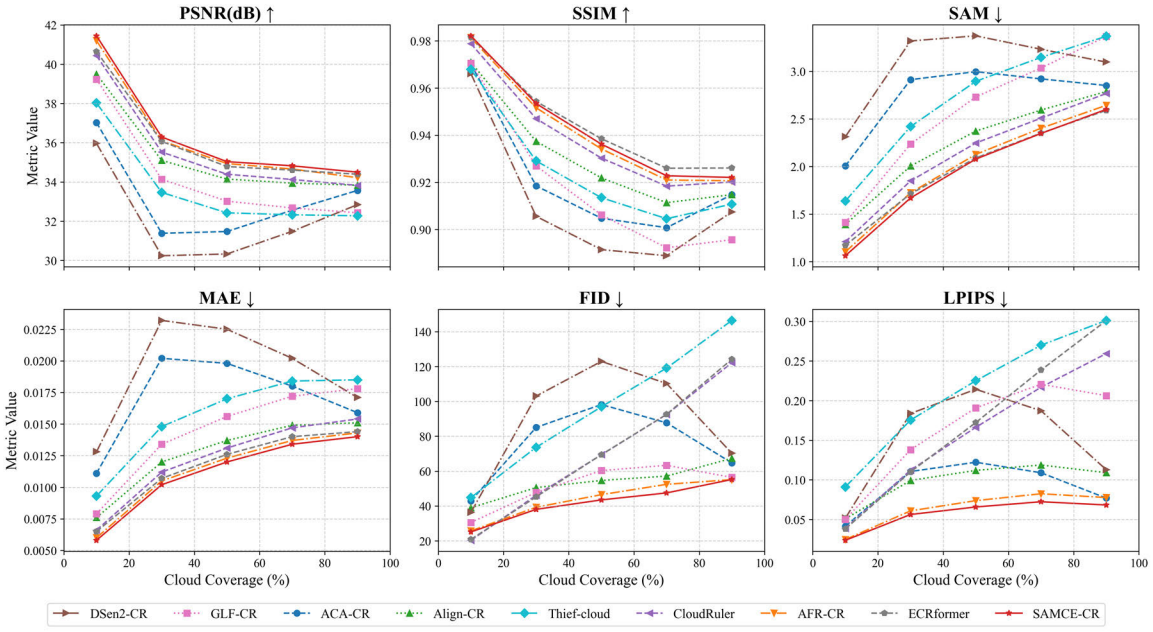


Fig. 8. Results of quantitative metrics across different cloud cover levels.

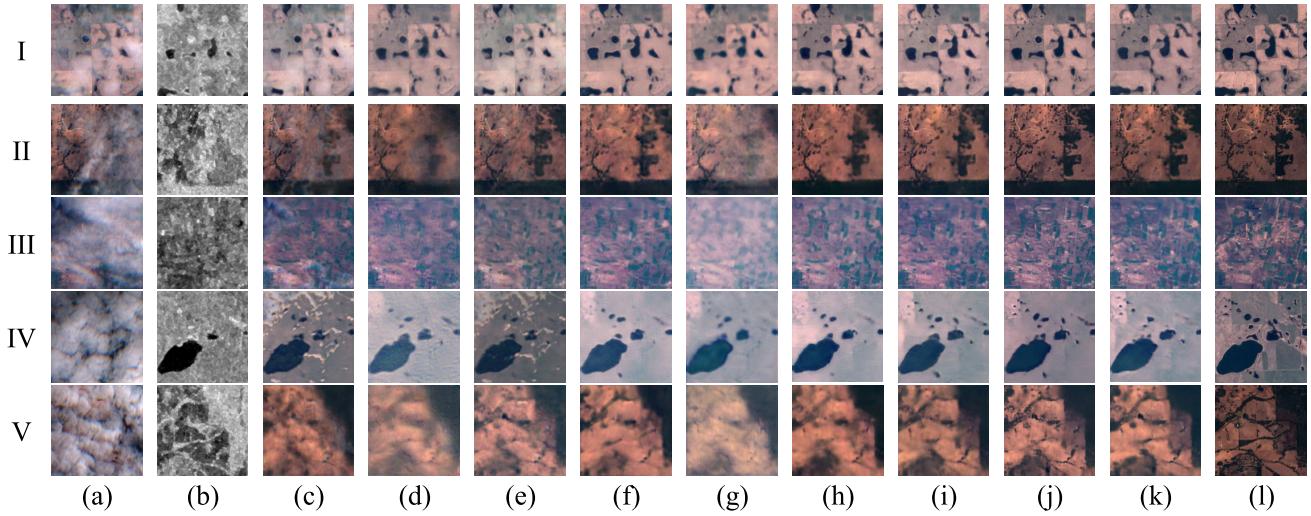


Fig. 9. Visual results for distinct cloud cover levels. (a) Cloudy image. (b) SAR image. (c) DSen2-CR. (d) GLF-CR. (e) ACA-CR. (f) Align-CR. (g) ThiefCloud. (h) CloudRuler. (i) AFR-CR. (j) ECRformer. (k) SAMCE-CR. (l) Cloud-free. Panels I–V correspond to cloud cover ranges of 0%–20%, 20%–40%, 40%–60%, 60%–80%, and 80%–100%, respectively.

objectively underscores its deficiencies in detail reconstruction and noise suppression within complex scenes. In summary, unlike the numerical fluctuations observed among different indicators when employing the comparison methods, the proposed method demonstrates consistently lower error rates and higher similarity levels across all evaluation items.

C. Performance Analysis Under Varying Cloud Cover Levels

To test the stability of our method under different weather conditions, we divided the test images from the LuojiaSET-OSFCR dataset into five groups based on their cloud coverage percentages (from 0%–20% to 80%–100%). Fig. 8 shows the quantitative results for these groups.

As expected, the performance of all evaluated models drops when the cloud cover becomes thicker. This happens because thick clouds block most of the optical information. For thin clouds (0%–20%), SAMCE-CR achieves highly competitive results. While the CloudRuler method obtains the lowest FID score, our proposed network secures the best scores in most other metrics, including an SSIM of 0.9821, an MAE of 0.0058, and a PSNR of 41.4412 dB. The advantage of our model becomes much clearer in thick cloud conditions. For example, in the extreme 80%–100% cloud coverage group, the PSNR of the ThiefCloud method falls sharply to 32.2732 dB, and its FID rises to 146.4492. Under the same challenging conditions, SAMCE-CR still reaches a PSNR of 34.4956 dB and limits the FID to 55.1867. This stability mainly comes

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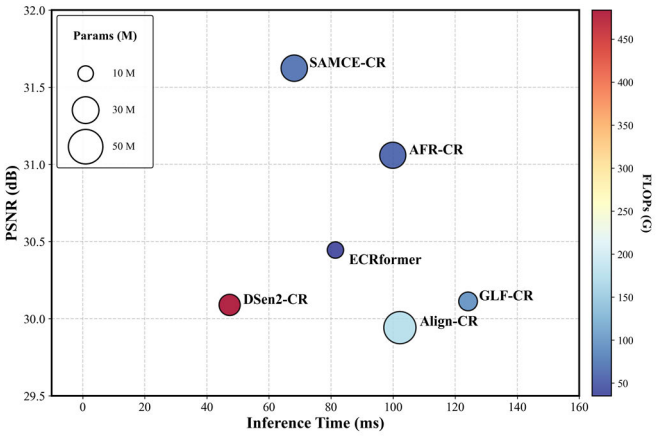


Fig. 10. Comparison of parameter count, FLOPs, inference time, and PSNR across different models in the M3R-CR Dataset. The x- and y-axes represent inference time and PSNR, respectively. The size of the bubbles denotes the Params, and the color indicates the FLOPs. A better model should be positioned toward the top-left corner with a smaller bubble size.

from our SAR structural prior and the multifrequency collaborative enhancement module. Because SAR signals can penetrate clouds, they provide a reliable geometric structure to help the network reconstruct the missing optical details, mitigating the performance drop caused by severe optical occlusion.

Furthermore, visual comparisons are provided in Fig. 9. Most baseline methods can remove thin clouds successfully. However, when dealing with thick clouds, their outputs often show obvious blurring and incorrect colors. In comparison, the images produced by SAMCE-CR demonstrate greater resilience. Although some details are inevitably lost in fully cloud-covered areas, the proposed method suffers much less degradation, maintaining relatively clear boundaries and more accurate textures than the competing baseline methods.

D. Model Complexity and Performance Analysis

To ensure that the evaluation of model complexity and performance bears practical relevance, this section selects mainstream de-clouding methods that likewise adhere to the multimodal fusion architecture as comparative baselines. As shown in Fig. 10, this article presents a comprehensive evaluation of these models across four metrics: Params, floating-point operations (FLOPs), the inference latency time (measured with a batch size of 1 on an NVIDIA RTX A4000 GPU), and PSNR. The experimental results indicate that, despite achieving the shortest inference time of 47.34 ms, the DSen2-CR model incurs substantial computational overhead, with FLOPs reaching 483.77 G, while its reconstruction quality remains relatively constrained, as evidenced by a PSNR of merely 30.0888 dB. In contrast, ECRformer, despite having the smallest parameter count of merely 11.28 M, is hindered by an inference time of 81.43 ms. The proposed method achieves an optimal balance between efficiency and performance, with a computational cost of merely 69.69 GFLOPs and an inference time of 68.15 ms, which significantly outperforms the comparison models GLF-CR, Align-CR, and AFR-CR in terms of inference speed. More importantly, while maintaining high efficiency, the proposed method achieves the highest PSNR value of 31.6229 dB, thereby demonstrating its superiority in cloud removal tasks.

E. Ablation Experiment and Analysis

To verify the efficacy of the proposed modules and their contributions to the overall performance, a comprehensive ablation study was conducted. To systematically assess the independent contribution and synergistic gain of MSPM, DCAM, and MDSGM in de-clouding tasks, seven variants (V0–V6) and the complete model (SAMCE-CR) were established for experimental evaluation. V0 serves as the performance benchmark without any proposed module; V1-V3 introduce a single module individually to validate its intrinsic efficacy; V4-V6 investigate the complementarity and fusion potential between modules through pairwise combinations; and the final SAMCE-CR model integrates all three components to demonstrate the optimality of the overall architecture. By comparing the performance of each variant, the contribution of individual modules to the enhancement of cloud removal efficacy utilizing SAR imagery can be elucidated.

1) Analysis of the Ablation Experiment Results for the M3R-CR Dataset: Fig. 11 and Table III present the qualitative and quantitative comparison results of ablation experiments on the M3R-CR dataset, respectively. Comprehensive analyses demonstrate that models lacking critical components encounter significant challenges in reconciling structural fidelity with spectral fidelity. The benchmark model V0 exhibits significant visual blur and distortion, whereas the complete model SAMCE-CR not only improves SSIM by 0.0103–0.9128 but also reduces SAM and MAE by 0.6965 and 0.0017, respectively, relative to V0. Single-module variants such as V1-V3 are constrained by limitations including texture smoothing, geometric distortion, and spectral shift; moreover, significant discrepancies evident in the residual maps indicate substantial deviations from ground truth values. On the core reconstruction index, the complete model achieves a PSNR value of 31.6229 dB, representing significant improvements of 0.5309, 0.6264, and 0.8261 dB relative to V1, V2, and V3, respectively. In addition, despite the improved visual integrity of V4–V6, the restoration of high-frequency details at building edges remains insufficient. Quantitative data also corroborate this observation. V5's lower SSIM highlights the pivotal function of DCAM in preserving spatial structural fidelity, whereas the enhanced performance of the complete model relative to V4 proves the indispensability of MDSGM in optimizing perceptual metrics such as LPIPS and reconstructing high-frequency details. Ultimately, the SAMCE-CR model achieves comprehensive optimality across all evaluation metrics. The uniformly dark blue tone exhibited by its residual map intuitively demonstrates an exceptionally low reconstruction error, thereby proving the design efficacy of the synergistic integration of MSPM, DCAM, and MDSGM.

2) Analysis of the Ablation Experiment Results for the LuojiaSET-OSFCR Dataset: Fig. 12 and Table IV illustrate the qualitative and quantitative outcomes of the ablation

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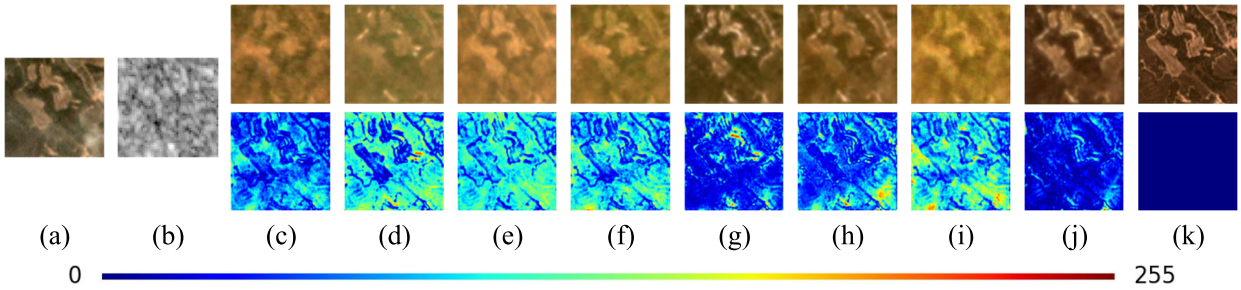


Fig. 11. Ablation experimental qualitative results of the M3R-CR dataset. (a) Cloudy image. (b) SAR image. (c) V0. (d) V1. (e) V2. (f) V3. (g) V4. (h) V5. (i) V6. (j) SAMCE-CR. (k) Cloud-free.

TABLE III
QUALITATIVE RESULTS OF ABLATION EXPERIMENT OF THE M3R-CR DATASET

Method	MSPM	DCAM	MDSGM	PSNR↑	SSIM↑	SAM↓	MAE↓	FID↓	LPIPS↓
V0				30.8414	0.9025	6.1335	0.0246	40.4199	0.1086
V1	√			31.0920	0.9089	5.7897	0.0242	35.9207	0.0931
V2		√		30.9965	0.9085	5.5360	0.0243	40.2704	0.1026
V3			√	30.7968	0.9056	5.8855	0.0252	39.7907	0.1043
V4	√	√		31.4216	0.9126	**5.4231**	0.0230	37.9551	0.0955
V5	√		√	31.5301	0.9081	5.7038	0.0230	39.2987	0.0917
V6		√	√	31.2078	0.9081	5.7231	0.0237	39.3284	0.1029
SAMCE-CR	√	√	√	**31.6229**	**0.9128**	5.4370	**0.0229**	**35.4971**	**0.0901**

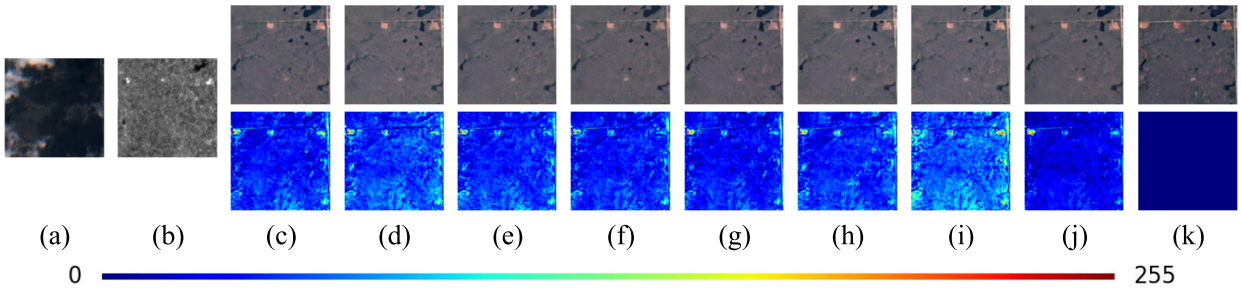


Fig. 12. Ablation experimental qualitative results of the LuojiaSET-OSFCR dataset. (a) Cloudy image. (b) SAR image. (c) V0. (d) V1. (e) V2. (f) V3. (g) V4. (h) V5. (i) V6. (j) SAMCE-CR. (k) Cloud-free.

TABLE IV
QUALITATIVE RESULTS OF ABLATION EXPERIMENT OF THE LUOJIASET-OSFCR DATASET

Method	MSPM	DCAM	MDSGM	PSNR↑	SSIM↑	SAM↓	MAE↓	FID↓	LPIPS↓
V0				36.1394	0.9420	2.0132	0.0114	35.4249	0.0622
V1	√			36.2763	0.9419	1.9951	0.0113	33.8402	0.0629
V2		√		36.2358	0.9419	2.0091	0.0114	**32.5950**	0.0592
V3			√	36.1818	0.9414	2.0179	0.0114	35.2554	0.0659
V4	√	√		36.2322	0.9418	2.0069	0.0114	35.1199	0.0622
V5	√		√	36.3065	0.9422	1.9892	0.0112	33.8773	0.0608
V6		√	√	36.2774	0.9422	2.0027	0.0113	33.3501	0.0590
SAMCE-CR	√	√	√	**36.4149**	**0.9433**	**1.9510**	**0.0111**	33.3958	**0.0572**

study performed on the LuojiaSET-OSFCR dataset, respectively. The complete model not only successfully eliminates cloud artifacts visually and exhibits the highest ground object fidelity but also attains optimal performance across core quantitative evaluation metrics; specifically, the PSNR is as high as 36.4149 dB, the SSIM is improved to 0.9433, and the LPIPS is reduced to 0.0572. Concurrently, the uniform dark blue tone exhibited by its residual map serves as an

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TABLE V
COMPUTATIONAL COST OF ABLATION VARIANTS

Method	M3R-CR Dataset		
	Param (M)	FLOPs (G)	Inference Time (ms)
V0	26.10	55.13	62.86
V1	26.46	65.29	65.37
V2	26.10	55.15	61.38
V3	28.85	59.47	67.05
V4	26.47	65.35	68.09
V5	29.22	69.63	73.08
V6	28.85	59.47	70.90
SAMCE-CR	29.22	69.69	68.15
Method	LuojiaSET -OSFCR Dataset		
	Param (M)	FLOPs (G)	Inference Time (ms)
V0	26.11	55.43	62.49
V1	26.48	65.58	66.93
V2	26.11	55.48	67.03
V3	28.86	59.77	74.91
V4	26.48	65.80	74.45
V5	29.23	69.93	74.89
V6	28.87	59.83	73.97
SAMCE-CR	29.23	70.15	73.55

intuitive indicator of low reconstruction error. Conversely, models lacking core components exhibit varying degrees of image distortion and metric degradation. Baseline V0, with all proposed modules removed, exhibits the most significant performance drop, with the PSNR declining to 36.1394 dB. Single-module variants (V1–V3) exhibit pronounced spectral shifts and texture smoothing upon reconstruction, with bright green regions in the residual maps indicating great deviations from the ground truth. Although the dual-module combinations (V4–V6) enhance the integrity of spatial structures, they remain insufficient for recovering high-frequency details at the edges of ground objects. Quantitative comparisons further corroborate the aforementioned visual limitations. Relative to V4 (without MDSGM) and V6 (without MSPM), the complete model achieves optimization of SAM and MAE errors to 1.9510 and 0.0111, respectively. Although variant V2 shows a single advantage in terms of the FID index, the comprehensive leading position of all data from the complete model fully proves that the three modules have an indispensable synergistic gains effect in improving global perception quality of images and suppressing spectral distortion.

3) Computational Cost and Synergistic Analysis: To investigate the cost efficiency, functional redundancy, and synergistic gains of the proposed modules, Table V reports their incremental computational costs. These include Params, FLOPs, and inference time.

The complete SAMCE-CR model achieves significant improvements with a total Params increment of only 3.12 M and a marginal inference time increase of roughly 5–11 ms over V0. Specifically, MSPM needs about 0.36 M Params to extract multiscale representations. Notably, DCAM introduces a minimal Params overhead of about 0.01 M. It employs lightweight convolutional layers to compute spatial offsets for feature warping rather than heavy feature extraction networks. Although it slightly increases FLOPs by 0.05G and brings a marginal rise in inference time, this compact design preserves structural details without noticeably increasing model size and latency. Comparing V1, V4, and SAMCE-CR explains their synergistic necessity. In Table IV, Variant V4 shows a slight PSNR drop compared to V1. This reflects a compromise rather than functional redundancy. V1 solely uses MSPM, generating overly smoothed images for higher pixel-level scores. Introducing DCAM in V4 adds geometric constraints to align optical features with SAR structures, which naturally reduces pixel-level smoothness. Because V4 lacks a mechanism to balance these features, performance drops slightly. Finally, MDSGM acts as a dynamic gate to optimally fuse these complementary components. The best performance of SAMCE-CR confirms these modules have no functional overlap and work together effectively.

TABLE VI
SENSITIVITY ANALYSIS OF THE SPATIAL KERNEL SIZE K

	PSNR	SSIM	SAM	MAE	FID	LPIPS
K=2	31.5367	0.9144	5.2196	0.0229	37.5601	0.0912
K=3	31.6229	0.9128	5.4370	0.0229	35.4971	0.0901
K=4	31.4127	0.9115	5.8122	0.0229	37.2713	0.0935

4) Sensitivity Analysis of Deformable Sampling Points: To validate the rationale for selecting exactly nine sampling points, which corresponds to a spatial kernel size of $K = 3$. In the DCAM, we conducted a sensitivity analysis comparing $K = 2$, 3, and 4. As shown in Table VI, the restoration performance exhibits distinct variations with different numbers of sampling points. Although $K = 2$ shows a slight numerical advantage with an SSIM of 0.9144 and a SAM of 5.2196, the proposed $K = 3$ configuration achieves the optimal balance between overall fidelity and perceptual quality. Specifically, $K = 3$ yields the best results across three core metrics, achieving a PSNR of 31.6229 dB, an LPIPS of 0.0901, and an FID of 35.4971. The significant reduction in FID, in particular, indicates that nine sampling points can more effectively capture complex land cover texture distributions, generating optical features with higher visual realism. This phenomenon occurs because a smaller sampling scope of $K = 2$ lacks sufficient contextual information to accurately calculate complex nonlinear deformations, whereas an excessively large receptive field of $K = 4$ tends to introduce irrelevant distant background noise outside the cloudy area that disturbs fine-grained cross-modal mapping. Therefore, setting $K = 3$ provides adequate alignment clues without injecting cross-modal interference, proving to be the optimal choice that effectively balances cross-modal alignment accuracy and visual perceptual quality.

5) Qualitative Spectral Analysis and Interpretability: To explicitly validate the physical interpretability of the adaptive frequency decoupling mechanism defined in (10), we conduct a qualitative spectral analysis on the predicted decoupling boundary. As established, the coefficient 0.5 strictly constrains the search space within the valid physical spectrum up to the Nyquist limit [36]. Through data-driven training, the network autonomously learns to dynamically position the decoupling boundary based on the severity of cloud occlusions.

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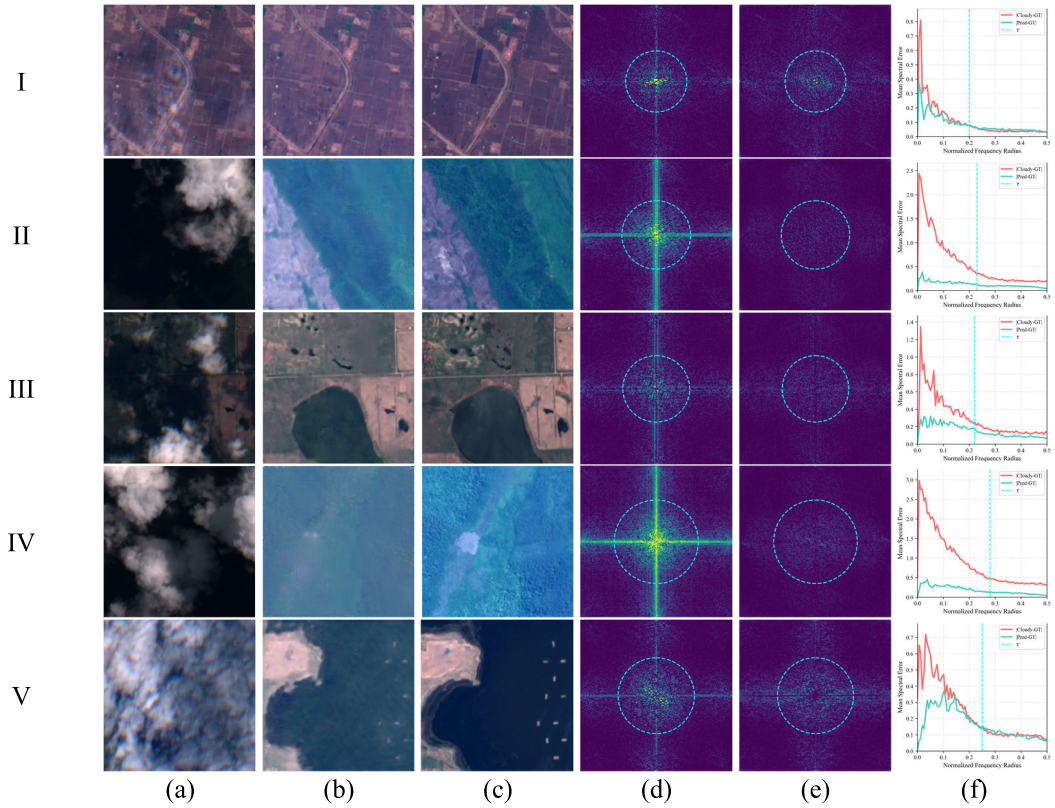


Fig. 13. Spectral analysis of the adaptive frequency decoupling mechanism under increasing cloud cover levels. Rows from top to bottom are arranged in ascending order of cloud occlusion severity. (a)–(c) Cloudy input image, predicted result, and cloud-free ground truth. (d)–(e) FFT magnitude difference maps of |Cloudy−GT| and |Prediction−GT|, respectively. (f) Corresponding radial error profiles of spectral differences. The cyan dashed circles indicate the dynamically predicted decoupling boundaries.

To illustrate this, Fig. 13 visualizes the spectral differences between the cloudy/predicted images and the cloud-free ground truth, alongside their corresponding radial error profiles, across representative cloud conditions with varying coverage and thickness. The bandwidth of low-frequency degradation is determined not only by cloud cover extent, but also critically by cloud thickness. Scenarios with optically thick clouds II and IV exhibit wider spectral degradation and correspondingly larger decoupling boundaries, even when cloud coverage is partial. In contrast, scenes with extensive but thin clouds show relatively narrower degradation bands, and the learned boundary adjusts accordingly. The radial error profiles further demonstrate that spectral errors are predominantly concentrated within this dynamically learned boundary and are effectively suppressed after restoration. Without empirical manual tuning, the module isolates low-frequency degradation via this adaptive physical boundary, endowing the binary decoupling strategy with physical interpretability and adaptability to diverse cloud types.

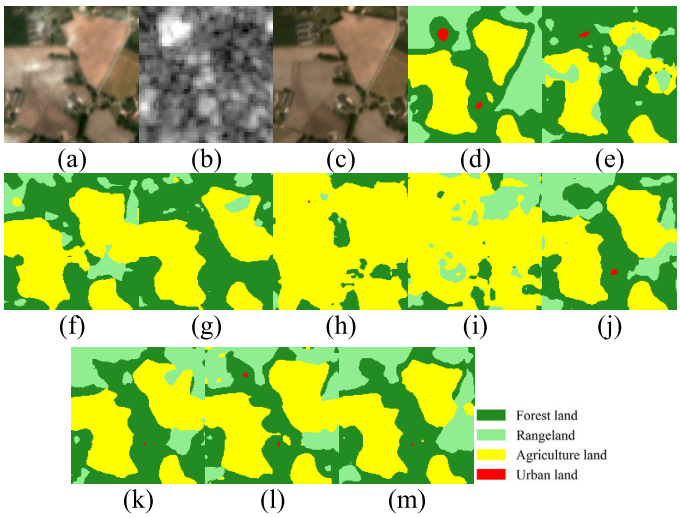


Fig. 14. Land use classification assessment of the M3R-CR dataset. (a) Cloudy image. (b) SAR image. (c) Cloud-free. (d) Cloud-free-LC. (e) DSen2-CR. (f) GLF-CR. (g) ACA-CR. (h) Align-CR. (i) ThiefCloud. (j) CloudRuler. (k) AFR-CR. (l) ECRformer. (m) SAMCE-CR.

F. Quality Assessment of Semantic Recovery

To verify the effectiveness of the cloud removal model in restoring the semantic structure of the land surface, this article conducts a semantic segmentation evaluation experiment using the DeepLabV3 + model based on ResNet-50 on the pixel-level land cover labels produced by the M3R-CR dataset. By feeding the predicted cloud-removed images into the segmentation network trained on real cloud-free imagery, we compute four evaluation metrics: overall accuracy (OA), mean Accuracy (mAcc), frequency-weighted accuracy (FWA), and mean intersection over union (mIoU). Because land cover datasets often present class imbalance,

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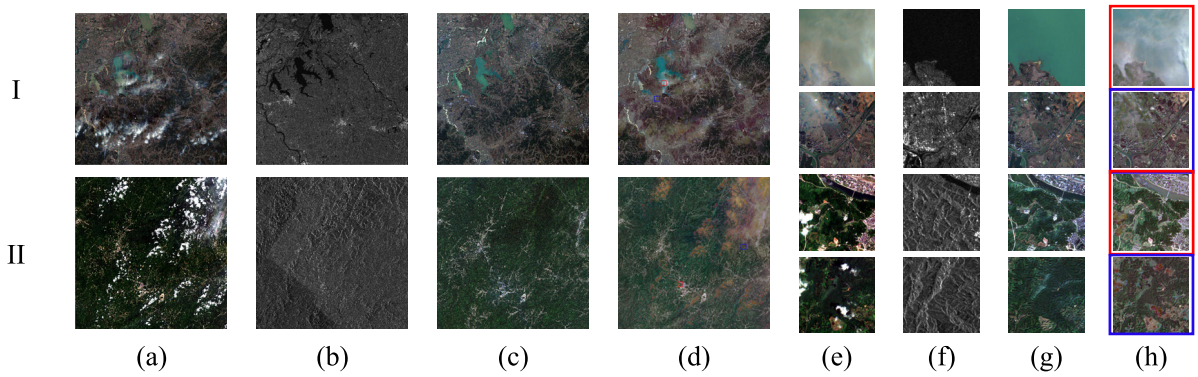


Fig. 15. Results of cloud removal for two different large-scale scenarios. Scenario I is from the Poyang Lake basin, whilst Scenario II is from Wuyi Mountain National Park. (a)–(d) Cloud-covered optical image, the SAR image, the cloud-free optical image, and the results obtained using the SAMCE-CR model, respectively. (e)–(h) Detailed views of the areas enclosed by the red and blue rectangles in the corresponding images on the left-hand side.

TABLE VII
QUANTITATIVE RESULTS OF LAND USE CLASSIFICATION ON THE M3R-CR DATASET

	OA(%) ↑	mAcc (%) ↑	FWA (%) ↑	MIoU (%) ↑
DSen2-CR	61.44	36.42	44.42	27.08
GLF-CR	52.90	42.09	35.08	26.81
ACA-CR	63.59	40.82	47.17	31.31
Align-CR	50.07	36.04	31.80	22.10
ThiefCloud	39.30	29.37	22.04	16.38
CloudRuler	67.07	50.48	50.82	39.95
AFR-CR	64.64	44.80	48.33	34.30
ECRformer	65.24	50.02	49.50	39.50
SAMCE-CR	**68.02**	**52.14**	**51.80**	**41.31**

relying solely on mIoU may overlook the performance on minority classes. Therefore, mIoU is complemented by OA for global correctness, mAcc for unweighted category-level accuracy, and FWA for frequency- weighted performance. Together, these metrics quantitatively assess the semantic consistency between the generated images and the actual surface, providing a comprehensive evaluation of the model's capability to preserve land cover information. Fig. 14 and Table VII present the qualitative and quantitative assessment results of land use classification for the M3R-CR dataset, respectively.

As evidenced by Fig. 14 and Table VII, the synthesis of feature classification maps and quantitative indicators reveals that SAMCE-CR delivers the best performance in downstream surface classification subsequent to cloud removal. Specifically, the proposed method achieves an OA of 68.02% and an MIoU of 41.31%, representing objective improvements of 0.95% and 1.36% over the suboptimal model, CloudRuler, respectively. In contrast, ThiefCloud exhibits the poorest classification performance, achieving an OA of merely 39.30% and an MIoU as low as 16.38%. This degradation primarily stems from the method's inability to effectively suppress the speckle noise inherent to SAR imagery, which consequently leads to large-scale misclassification and severely blurred boundaries of ground objects. In addition, methods such as DSen2-CR and Align-CR exhibit a loss of high-frequency textural information during the reconstruction process, consequently yielding over-smoothed outputs and the omission of small-scale urban lands and fragmented grassland patches. However, this method not only effectively overcomes the aforementioned noise interference and texture loss problems but also accurately restores the complex, interleaved patterns of cultivated land and forestland in the core area, demonstrating significant advantages in maintaining the semantic consistency and structural integrity of ground objects.

G. Large-Scale Scenario Applications

In addition, to verify the generalization capability and practical application potential of the proposed network, this study selects local imagery from the Poyang Lake Basin and Wuyi Mountain National Park, which are representative regions characterized by cloudy and rainy climatic conditions in southern China, as the research subjects, and evaluates the network on large-scale scenario images with a size of 10 000 × 10 000 pixels. Given the inherent memory constraints of computing hardware, this study employs a sliding window strategy to process the entire image. First, the image is cropped into 600 × 600 pixels sub-blocks with an overlapping region of 300 pixels between adjacent blocks. Subsequently, the trained model is employed to perform inference on each sub-block individually. Finally, the inference results are seamlessly merged into a complete large-scale cloud-removed image through pixel averaging in the overlapping regions.

Fig. 15 shows the processing effect of SAMCE-CR in large-scale scenarios. Specifically, Scenario I includes buildings and aquatic bodies, with some areas covered by thin clouds. From the restoration results, it is evident that the proposed method effectively eliminates occlusion from most thin clouds and reconstructs the fundamental contours of water boundaries and buildings with clarity. Concurrently, owing to the overlapping mean stitching strategy, no significant visual stitching gaps or blocking artifacts are observed throughout the entire large-scale image. Scenario II encompasses complex surface coverage types, such as deep mountain forest land. The results demonstrate that the proposed method successfully eliminates cloud cover over forested areas. Leveraging the structural prior information derived from SAR imagery, the method

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reconstructs the undulating morphology and topographic texture of the ridgelines to a considerable extent. However, objectively speaking, spectral discrepancies persist between the reconstructed results and the ground-truth imagery in regions previously subjected to severe thick-cloud occlusion. Although limitations in color fidelity persist, the results further corroborate that the proposed method effectively preserves the fundamental spatial structure of ground objects under adverse observation conditions, thereby precluding complete feature blurring or loss. Therefore, the algorithm demonstrates favorable adaptability in handling large-scale cloudy scenarios and effectively accommodates diverse cloud types and complex land surface coverage.

VI. Conclusion

Efficient integration of SAR and multispectral imagery for cloud removal constitutes the cornerstone of ensuring all-weather Earth observation and seamless spatiotemporal monitoring. However, achieving accurate information fusion across cross-modal heterogeneous data remains challenging, which has long constrained the structural fidelity of cloudless reconstructed images and limited their application potential in downstream interpretation tasks. Therefore, this article proposes the SAMCE-CR method, which utilizes SAR-guided alignment and multifrequency domain co-enhancement techniques to facilitate the recovery of optical images occluded by clouds. Specifically, this study sequentially extracts high-quality SAR structural priors via MSPM, achieves cross-modal adaptive geometric alignment through DCAM, and differentially recovers high- and low-frequency details by integrating MDSGM, thereby synergistically enabling high-quality cloud removal with clear edges and preserved texture fidelity. The results of multidataset comparisons, ablation experiments, downstream semantic evaluation, and large-scale scenario applications demonstrate that the proposed method stably outperforms existing baselines across most metrics. This indicates that the proposed framework demonstrates superior environmental adaptability and structural reconstruction robustness when confronted with substantial modal disparities and complex cloud occlusion conditions. Due to the extreme complexity of cloud degradation and surface structures in real scenarios, it remains imperative to expand the boundary of multisource information fusion to achieve all-weather and high-fidelity remote sensing image restoration.

To this end, future research endeavors will concentrate on the incorporation of richer multimodal data and the exploration of advanced network architectures, aiming to construct a more robust and versatile cloud removal model for remote sensing imagery.

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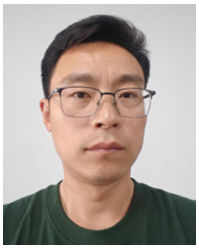
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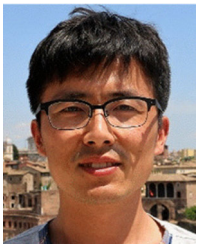
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